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Altium Schematic Instructions — Comms Board

Design Summary

This board implements the CubeSat's full RF communications system on a single PCB: a 437 MHz BPSK downlink transmitter (70cm) and a 145.9 MHz AFSK uplink receiver (2m), with an RP2040 (Pico module) as the digital controller.

CSKB Stack Connectors (from EPS, H1 signals + H2 power)

- |
- ├─ +3V3 → Si5351A, 74LVC1G86, MCP6022, RP2040
- ├─ +5V → PSA4-5043+ LNA, ADL5602 MMIC
- ├─ GND
- ├─ I2C_SCL/SDA → Internal Housekeeping Unit bus
- └─ VBAT (unused on comms, monitor only)

Si5351A (3-output clock gen, I2C controlled by RP2040)

- ├─ CLK0 (145.67 MHz) → XOR BPSK Mod → Tripler → BPF → ADL5602 MMIC → Output BPF → TX SMA
- ├─ CLK1 (145.9 MHz) → LO LPF → ADE-1+ Mixer LO input
- └─ CLK2 (spare)

Antenna → RX SMA → 2m BPF → PSA4 LNA → ADE-1+ Mixer → Sallen-Key LPF → MCP6022

Key design decisions: 1. **Power comes from the EPS via the CSKB stack connectors** — no on-board voltage regulators. +3V3 and +5V are generated on the EPS board and distributed through the stack connector. 2. **XOR BPSK modulation at 145.67 MHz** (before the tripler), not at 437 MHz. The 74LVC1G86 has massive timing margin at 146 MHz but fails at 437 MHz. BPSK phase states

are preserved through odd-order frequency multiplication ($3 \times 180^\circ = 540^\circ \equiv 180^\circ$). 3. **Frequency tripler** (class-C biased transistor) converts 145.67 MHz to 437 MHz. BFR92A (ft=5 GHz, SOT-23) for flight; 2N3904 for breadboard validation. 4. **Separate TX and RX SMA connectors** (no diplexer for prototype). 5. **Raspberry Pi Pico module** for prototype — bare RP2040 deferred to v2.

Altium Project Setup

1. Create a new PCB project: `Comms_Board.PrjPcb`
2. Add seven schematic sheets (flat design):
 - `Overview.SchDoc` — title sheet, block diagram, design notes (non-electrical)
 - `Clock_Gen.SchDoc` — Si5351A and I2C interface
 - `TX_Chain.SchDoc` — XOR modulator, tripler, BPF, MMIC
 - `RX_Chain.SchDoc` — PSA4-5043+ LNA, ADE-1+ passive mixer, LO LPF, IF termination, Sallen-Key LPF, MCP6022 gain stage
 - `Digital_Control.SchDoc` — RP2040 (Pico module), USB, GPIOs
 - `Power.SchDoc` — power distribution, bypass, and protection
 - `Connectors.SchDoc` — CSKB H1 + H2 stack connectors, SMA connectors, test headers
3. Each sheet uses A3 landscape format
4. Set the project's designator scope to "Flat" so R1, C1, etc. are unique across all sheets

Flat Design — How Inter-Sheet Connections Work

This project uses a **flat** (non-hierarchical) design, same as the EPS board. There is no top-level sheet with sheet symbols. Instead, nets connect between sheets using **net labels** and **power ports**:

- **Power ports** (+3V3, +5V, GND, VBAT) are global automatically — place the same power port symbol on any sheet and they connect.
- **Signal net labels** (I2C_SDA, CLK0_OUT, BPSK_DATA, etc.) are also global in a flat project — place the same net label on two different sheets and Altium connects them automatically.
- **No ports or sheet entries needed** — those are only for hierarchical designs. In a flat design, just use net labels with matching names across sheets.

This is simpler to set up and matches the EPS board's structure.

Net Name Convention

Use these global net names consistently across all sheets. Use **power port symbols** (not net labels) for supply rails so they are global automatically.

Net Name	Type	Description
+3V3	Power port	3.3V regulated rail (from EPS via bus)

Net Name	Type	Description
+5V	Power port	5.0V regulated rail (from EPS via bus)
GND	Power port	Ground (global)
VBAT	Power port	Unregulated battery bus from EPS (monitor only)
I2C_SDA	Net label	I2C data — Si5351A and internal housekeeping unit bus
I2C_SCL	Net label	I2C clock — Si5351A and internal housekeeping unit bus
BPSK_DATA	Net label	RP2040 GPIO to XOR gate data input
CLK0_OUT	Net label	Si5351A CLK0 output (145.67 MHz)
CLK1_OUT	Net label	Si5351A CLK1 output (145.9 MHz LO)
BPSK_145	Net label	XOR output, BPSK-modulated 145.67 MHz
TRIPLER_OUT	Net label	Tripler output, unfiltered 437 MHz
TX_437	Net label	Filtered 437 MHz, post-BPF
TX_OUT	Net label	MMIC output to TX SMA
RX_IN	Net label	RX SMA input to 2m BPF
RX_IF	Net label	ADE-1+ IF output (baseband, post-termination)
RX_FILT	Net label	Sallen-Key filter output
RX_BASEBAND	Net label	MCP6022 amplified output to Pico ADC

Note on Power Port Symbols Altium power port symbols come in several styles (circle, T-bar, arrow, etc.). The style is **cosmetic only** — the net name on the power port is what matters electrically. All power ports with the same net name connect globally regardless of symbol shape. Convention used in this design: - **T-bar** for supply rails (+3V3, +5V, VBAT) - **Circle with line** for GND - You can use any style as long as the net name matches exactly

Sheet 1: Overview

This is a non-electrical title/overview sheet. No components or nets are placed here — it serves as the front page of the schematic package and provides context for anyone reviewing or building the design.

Text Frame 1: Project Identification (Top of Sheet)

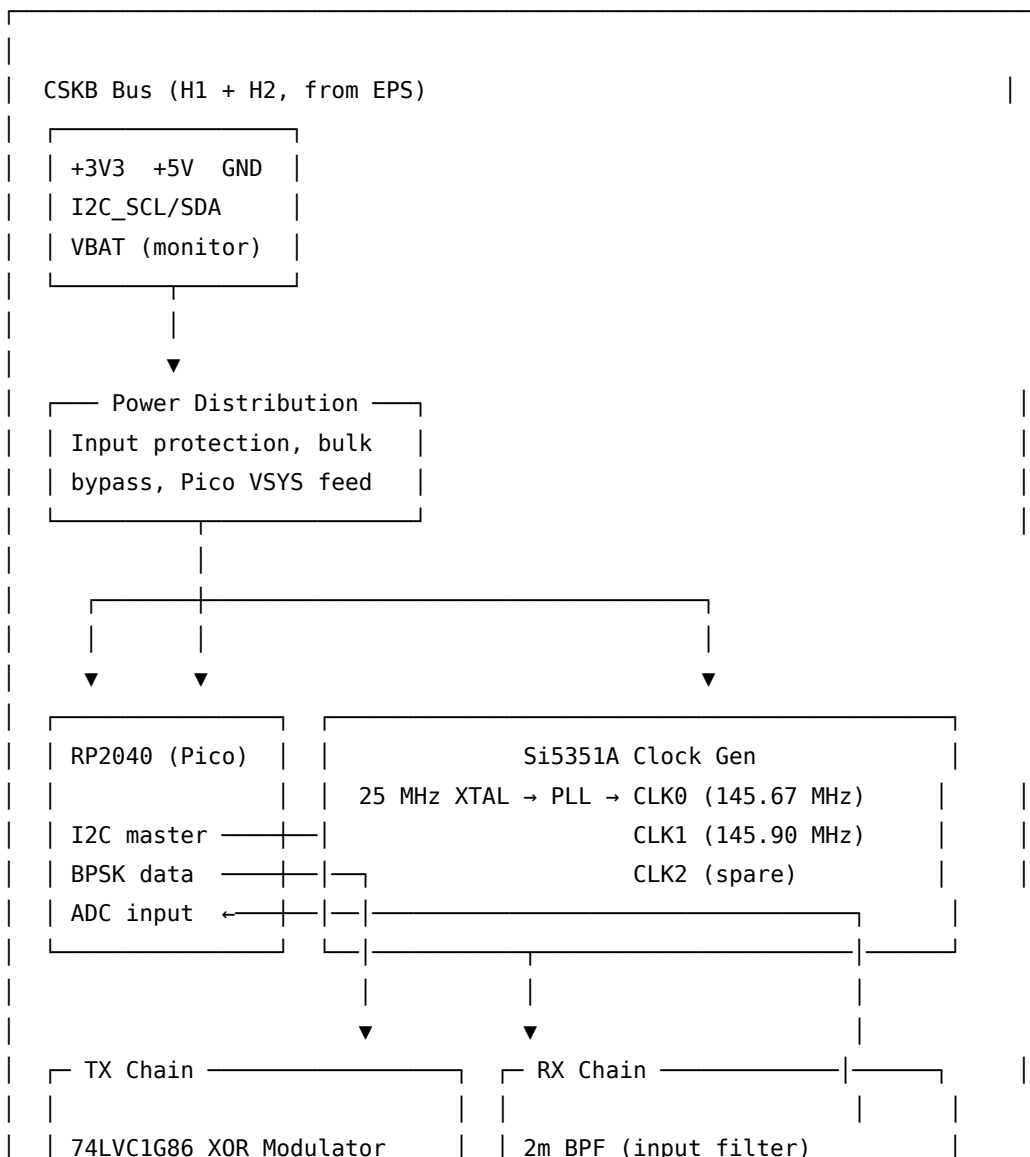
Place a large text frame across the top of the sheet:

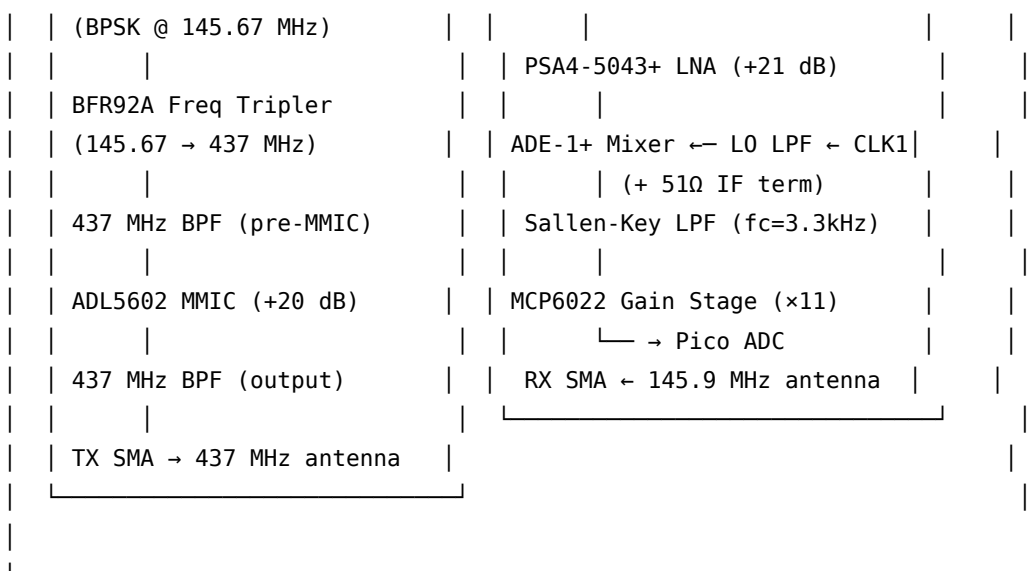
CubeSat Communications Board — 70cm BPSK TX / 2m AFSK RX

Project: CentiSat CubeSat — Senior Capstone Design Board: Communications Subsystem,
Single PCB Designer: Nick Grabbs / K15Y NG Revision: 0.2 Date: (today) Repository:
centisat/hardware/comms

Text Frame 2: System Block Diagram (Center of Sheet)

Place a text frame with the system architecture. In Altium, you can also paste a graphic (Place > Drawing Tools > Graphic or import a bitmap/SVG).





Text Frame 3: Power Rail Summary (Bottom-Left)

Power Rails

Rail	Source	Voltage	Consumers
+3V3	EPS via CSKB (H2.27/28)	3.3V $\pm 2\%$	RP2040, Si5351A, 74LVC1G86, MCP6022
+5V	EPS via CSKB (H2.25/26)	5.0V $\pm 2\%$	PSA4-5043+ LNA, ADL5602 MMIC, tripler bias, Pico VSYS
GND	EPS via CSKB (H2.29/30/32, H2.31 AGND)	0V	All
VBAT	EPS via CSKB (H2.45/46)	6.0–8.4V	Monitor only (not used for power)

Estimated current budget: +3V3: ~150 mA (RP2040 ~50mA, Si5351A ~30mA, logic ~10mA, op-amp ~10mA) +5V: ~140 mA (PSA4 LNA ~60mA, ADL5602 ~60mA, tripler ~5mA, Pico VSYS ~10mA, margin ~5mA) Total: <300 mA — well within EPS TPS62933F capacity (3A per rail). Note: PSA4 LNA bias can be GPIO-gated for RX duty-cycling.

Text Frame 4: Net Name Convention (Bottom-Center)

Net Name Convention

Power rails use **power port symbols** (global across all sheets). Signal nets use **net labels** (also global in flat design).

Net	Type	Description
+3V3, +5V, GND, VBAT	Power port	Supply rails (global)
I2C_SDA, I2C_SCL	Net label	I2C bus (Si5351A + IHU bus)
CLK0_OUT, CLK1_OUT	Net label	Si5351A clock outputs
BPSK_DATA	Net label	RP2040 → XOR modulator
TX_OUT, RX_IN	Net label	RF paths to SMA connectors
RX_BASEBAND	Net label	MCP6022 output → Pico ADC

Text Frame 5: Key Design Notes (Bottom-Right)

Key Design Decisions

1. Power from EPS via CSKB bus — no on-board regulators
2. BPSK modulation at 145.67 MHz (before tripler) — 74LVC1G86 fails at 437 MHz, see [overview.md](#) timing analysis
3. Odd-order frequency multiplication preserves BPSK phase states ($3 \times 180^\circ = 540^\circ \equiv 180^\circ \bmod 360^\circ$)
4. Tripler bias $47\text{k}\Omega$ — empirically optimized (-7.4 dBm @ 437 MHz)
5. All RF filter caps must be C0G/NP0 (not X7R)
6. I2C addresses: Si5351A = 0x60, EPS LTC4162 = 0x68 (no conflict)
7. Flat schematic design — nets connect by name across sheets
8. Prototype uses Pico module; bare RP2040 deferred to v2

Text Frame 6: Sheet Index (Right Edge)

Sheet Index

Sheet	Title	Contents
1	Overview	This sheet — block diagram, design notes
2	Clock Gen	Si5351A, 25 MHz crystal, I2C, clock outputs
3	TX Chain	XOR modulator, tripler, BPF, ADL5602 MMIC
4	RX Chain	PSA4 LNA, LO LPF, ADE-1+ mixer, IF term, Sallen-Key LPF, MCP6022 gain
5	Digital Control	RP2040 (Pico module), GPIOs, USB
6	Power	Distribution, protection, bypass
7	Connectors	CSKB H1 + H2 bus, SMA ports, test points

Text Frame 7: Revision History (Bottom, or Separate Table)

Revision History

Rev	Date	Author	Description
0.1	2026-04-10	NG	Initial schematic — all sheets placed, basic connectivity
0.2	2026-04-13	NG	Added Connectors sheet, stack bus, removed on-board regulators, added overview sheet
0.3	2026-04-15	NG	Added COMMS_IRQ on GP3 and SPI slave wiring on GP16–19; initial Pumpkin pin-number lock
0.4	2026-04-15	NG	Rebuilt connectors against Pumpkin datasheet Rev. E: bus renamed PC/104 → CubeSat Kit Bus (CSKB); pin numbers remapped H10 → H1/H2; connector family corrected to 2× Samtec ESQ-126-39-G-D stackthrough (2×26, 52 pins each); see system/interfaces/cskb_pinmap.md

How to Build This Sheet in Altium

1. Create a new schematic sheet: `Overview.SchDoc`
2. Use Place > Text Frame for each section above
3. For the block diagram, you can either:
 - Use text frames with monospace font (as shown above)
 - Use Place > Drawing Tools > Line/Rectangle to draw boxes and arrows
 - Import a PNG/SVG of the block diagram (Place > Drawing Tools > Graphic)
4. No electrical components, no nets, no power ports on this sheet
5. Set the title block: “CubeSat Comms Board — Overview”, Rev 0.2

Sheet 2: Clock Gen (Si5351A)

This sheet contains the Si5351A clock generator, its crystal, I2C pull-ups, and output coupling networks.

Text Frame (Place > Text Frame, top of sheet):

CLOCK GENERATOR — Si5351A (I2C address 0x60). Generates three independent clock outputs from a single 25 MHz crystal. CLK0 = 145.67 MHz (TX carrier, tripled to 437 MHz). CLK1 = 145.9 MHz (RX local oscillator for ADE-1+ passive mixer; see RX sheet). CLK2 = spare. All outputs driven at 8 mA (max) into 50Ω loads. Controlled by RP2040 over I2C.

Section A: Si5351A IC (Center of Sheet)

Component Search Manufacturer Part Search for “Si5351A-B-GT” (MSOP-10 package). Place in the center of the sheet.

Pin-by-Pin Connections

Pin	Name	Connect To	Notes
1	VDD	+3V3	Main supply, bypass with 100nF + 10μF
2	GND	GND	
3	CLK0	CLK0_OUT via 100nF DC block	TX carrier (145.67 MHz)
4	CLK1	CLK1_OUT via 100nF DC block	RX LO (145.9 MHz)
5	CLK2	No connect (or spare header)	Reserve for future use
6	VDDO	+3V3	Output driver supply, bypass with 100nF
7	SCL	I2C_SCL	I2C clock
8	SDA	I2C_SDA	I2C data
9	XA	Crystal pin 1	25 MHz crystal
10	XB	Crystal pin 2	25 MHz crystal

Bypass Capacitors

Ref	Value	Package	Connection	Purpose
C1	100nF, 16V, X7R	0402	VDD (pin 1) to GND	HF bypass
C2	10μF, 10V, X5R	0805	VDD (pin 1) to GND	Bulk bypass
C3	100nF, 16V, X7R	0402	VDDO (pin 6) to GND	Output driver bypass

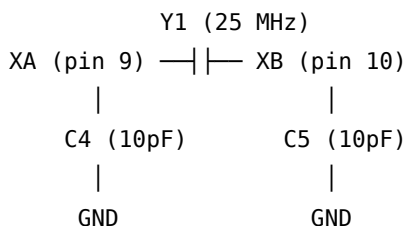
Note (Place > Note, next to bypass caps):

C1/C2 at VDD: place C1 (100nF) directly at pin 1 with shortest possible traces. C2 (10μF) can be further away. C3 at VDDO: place directly at pin 6. Per Skyworks AN619, poor VDDO bypass causes increased output jitter. Keep all bypass return paths to GND short.

Section B: 25 MHz Crystal (Left of Si5351A)

Components

Ref	Value	Package	Connection	Notes
Y1	25.000 MHz, 10pF load	HC-49/US or 3.2x2.5mm SMD	XA (pin 9) and XB (pin 10)	Fundamental mode
C4	10pF, 50V, C0G/NP0	0402	XA (pin 9) to GND	Crystal load cap
C5	10pF, 50V, C0G/NP0	0402	XB (pin 10) to GND	Crystal load cap



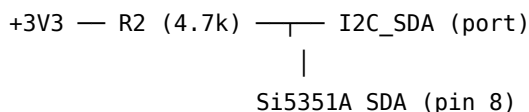
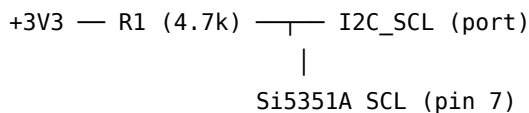
Note (Place > Note, next to crystal):

Crystal load capacitance: $CL = (C4 \times C5) / (C4 + C5) + C_{stray}$. With $C4=C5=10\text{pF}$ and $\sim 3\text{pF}$ stray: $CL \approx 8\text{pF}$. Match to crystal's specified load capacitance. If crystal specifies 8pF load, use 10pF caps. If crystal specifies 10pF load, use 15pF caps. C0G/NP0 dielectric is mandatory — X7R temperature drift will shift frequency.

Section C: I2C Pull-Up Resistors (Bottom-Left)

Components

Ref	Value	Package	Connection
R1	4.7k Ω , 1%	0402	I2C_SCL to +3V3
R2	4.7k Ω , 1%	0402	I2C_SDA to +3V3



Note (Place > Note, next to pull-ups):

I2C pull-ups: 4.7k Ω to +3V3. Si5351A address = 0x60 (7-bit). Standard mode (100 kHz) or Fast mode (400 kHz). These are the ONLY I2C pull-ups on the comms board — the EPS board has its own pull-ups for the LTC4162. If the bus is too slow (rounded edges on SCL), try 2.2k Ω . If multiple devices on the bus have pull-ups, the effective resistance drops — may need to remove one set.

Section D: Clock Output Coupling (Right of Si5351A)

Each clock output needs a DC blocking cap and optional series resistor for impedance matching.

Components

Ref	Value	Package	Connection	Purpose
C6	100nF, 16V, X7R	0402	CLK0 (pin 3) to CLK0_OUT net	DC blocking
R3	33 Ω , 1%	0402	Series in CLK0_OUT path	Source impedance matching
C7	100nF, 16V, X7R	0402	CLK1 (pin 4) to CLK1_OUT net	DC blocking
R4	33 Ω , 1%	0402	Series in CLK1_OUT path	Source impedance matching

CLK0 (pin 3) — C6 (100nF) — R3 (33 Ω) $\rightarrow$ CLK0_OUT (port)

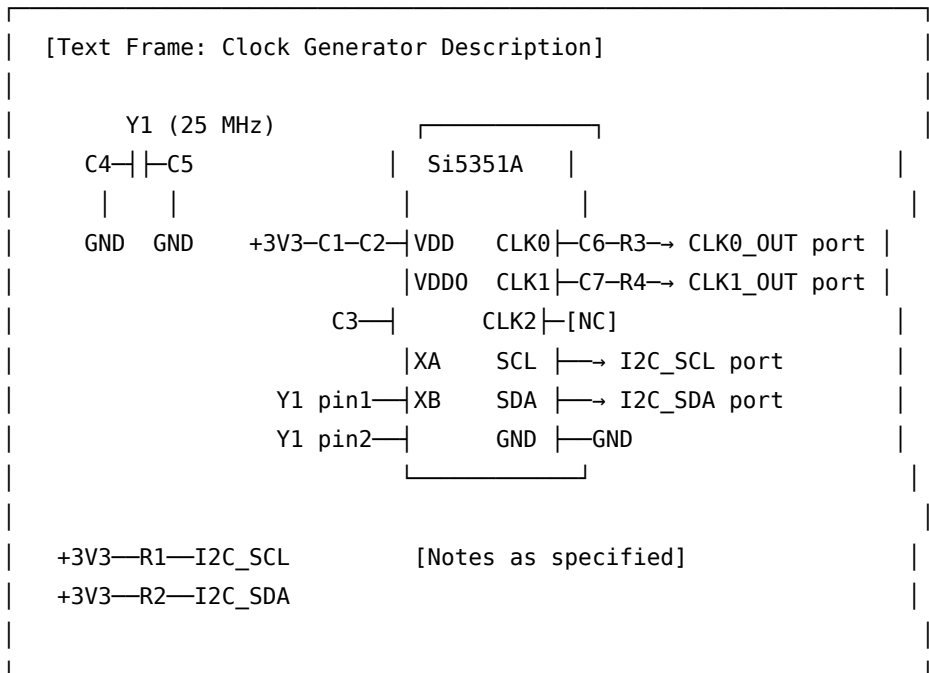
CLK1 (pin 4) — C7 (100nF) — R4 (33 Ω) $\rightarrow$ CLK1_OUT (port)

CLK2 (pin 5) — [no connect X] or spare header

Note (Place > Note, next to output coupling):

DC blocking caps prevent Si5351A output bias from affecting downstream stages. 33 Ω series resistors provide impedance matching and reduce ringing on traces. Si5351A outputs are push-pull CMOS with $\sim$ 50 Ω source impedance at 8mA drive setting. CLK0 feeds the XOR BPSK modulator at 145.67 MHz. CLK1 feeds the ADE-1+ passive mixer LO at 145.9 MHz (target +7 dBm at the mixer LO pin) through a 5-pole LC LPF on the RX sheet that suppresses square-wave harmonics. **Consider populating R4 as 0 Ω instead of 33 Ω** to preserve drive into the LPF/mixer chain — verify CLK1 power on the bench before committing. CLK2 reserved — bring to a test header if board space permits.

Sheet 2 Schematic Layout Suggestion



Sheet 2 Ports Checklist

- ☐ I2C_SDA — bidirectional, to RP2040 and CSKB H1.41
- ☐ I2C_SCL — bidirectional, to RP2040 and CSKB H1.43
- ☐ CLK0_OUT — output, 145.67 MHz to TX Chain
- ☐ CLK1_OUT — output, 145.9 MHz to RX Chain
- ☐ +3V3 — power port (global, from EPS via bus)
- ☐ GND — power port (global)

Sheet 3: TX Chain

This sheet implements the complete 437 MHz BPSK transmit chain: XOR modulator, frequency tripler, band-pass filter, and MMIC amplifier.

Text Frame (Place > Text Frame, top of sheet):

TX CHAIN — 437 MHz BPSK downlink transmitter. The carrier from the Si5351A (145.67 MHz) is BPSK-modulated by the 74LVC1G86 XOR gate, then frequency-tripled to 437 MHz by a class-C biased BFR92A transistor. A 3-pole LC BPF selects the 3rd harmonic and rejects the fundamental and 2nd harmonic. The ADL5602 MMIC provides ~20 dB gain to bring the signal to ~+10 dBm output. BPSK phase states are preserved through odd-order multiplication: $3 \times 180^\circ = 540^\circ \equiv 180^\circ$.

Section A: XOR BPSK Modulator (Left of Sheet)

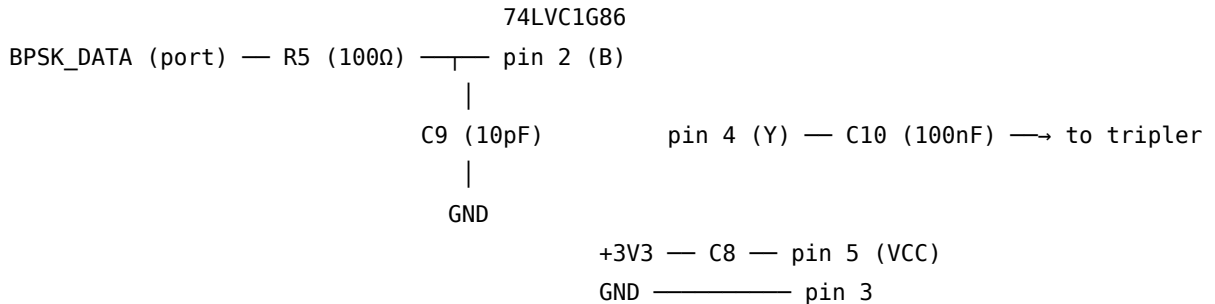
Components

Ref	Value/Part	Package	Connection
U1	SN74LVC1G86DBVR	SOT-23-5	XOR gate
C8	100nF, 16V, X7R	0402	VCC (pin 5) to GND bypass
R5	100Ω, 1%	0402	Series on BPSK_DATA line
C9	10pF, 50V, C0G	0402	BPSK_DATA to GND (edge softening)

Pin Connections

Pin	Name	Connect To	Notes
1	A	CLK0_OUT (from Si5351A)	145.67 MHz carrier input
2	B	BPSK_DATA via R5/C9 network	Data from RP2040 GPIO
3	GND	GND	
4	Y	BPSK_145 via 100nF DC block (C10)	Modulated output
5	VCC	+3V3	Bypass with C8

CLK0_OUT (port) → pin 1 (A)



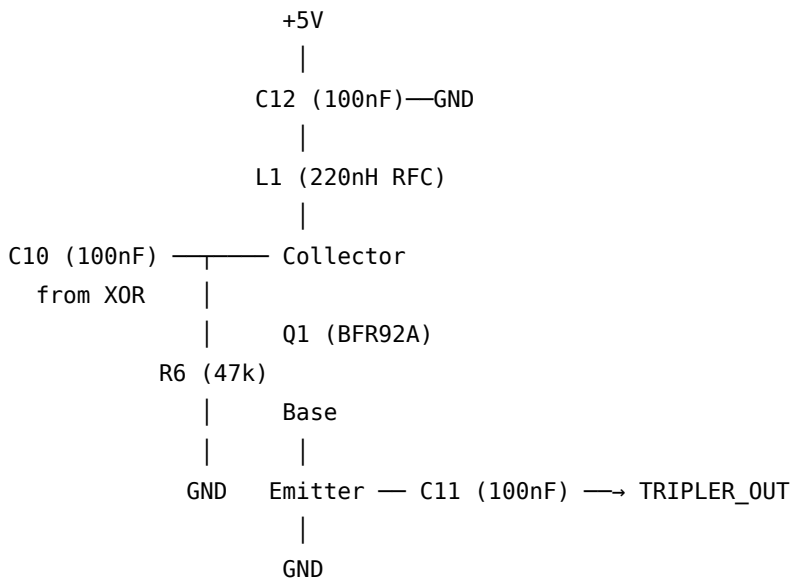
Note (Place > Note, next to XOR gate):

74LVC1G86 operates at 145.67 MHz where it has massive timing margin (half-period = 3.43 ns vs typical tpd = 0.6–2.3 ns). R5/C9 on the data input provide edge softening to reduce spectral splatter — the data rate is only 1200–9600 baud so the RC time constant (1 ns) has zero effect on data. The BPSK signal at 145.67 MHz feeds the frequency tripler; phase states (0°/180°) are preserved through the 3× multiplication. See overview.md for the full timing analysis showing why 437 MHz operation fails.

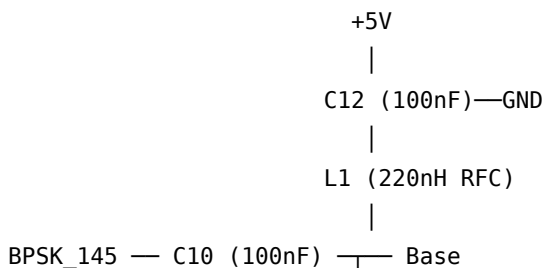
Section B: Frequency Tripler (Center-Left)

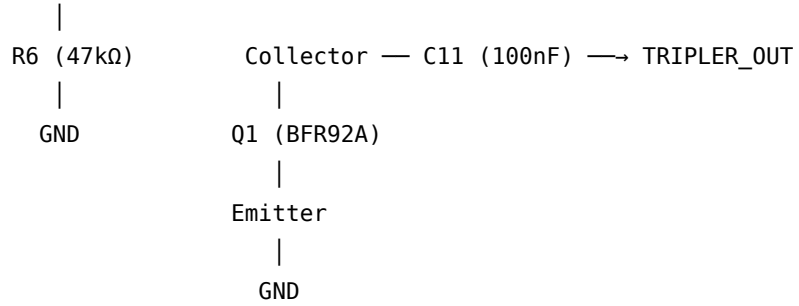
Components

Ref	Value/Part	Package	Connection	Notes
Q1	BFR92A (flight) / 2N3904 (proto)	SOT-23 / SOT-23	Tripler transistor	ft=5GHz / ft=300MHz
R6	47k Ω , 5%	0402	Base to GND	Class-C bias (optimized from breadboard testing)
L1	220nH, $\pm 5\%$	0805 (Coilcraft 0805CS)	VCC to collector	RFC, SRF > 1 GHz
C10	100nF, 16V, X7R	0402	XOR output to base	DC blocking / AC coupling input
C11	100nF, 16V, X7R	0402	Collector to BPF input	DC blocking / AC coupling output
C12	100nF, 16V, X7R	0402	+5V to GND at RFC top	Local bypass for tripler supply



Wait — the standard tripler topology has the output from the collector, not the emitter. Corrected circuit:





Note (Place > Note, next to tripler):

Class-C frequency tripler. R6 (47kΩ) biases the transistor below cutoff — it only conducts on positive input peaks, generating strong harmonic content. 47kΩ was empirically optimized on the breadboard prototype: -7.4 dBm at 437 MHz with 2N3904 (vs -12.2 dBm at 10kΩ). BFR92A (ft=5 GHz, SOT-23) is the flight transistor; 2N3904 (ft=300 MHz) validated during bring-up. RFC L1 provides DC bias path while presenting high impedance at RF. SRF of L1 must be well above 437 MHz — Coilcraft 0805CS-221XJRC (SRF ~1.5 GHz). Supply is +5V for maximum collector swing and harmonic output.

Section C: 437 MHz Band-Pass Filter (Center)

3-pole LC band-pass filter centered at 437 MHz. This is the most critical filter in the TX chain — it must reject the fundamental (145.67 MHz) and 2nd harmonic (291.3 MHz) by >30 dB.

Components

Ref	Value	Package	Dielectric	Notes
C13	10pF, 50V	0402	C0G/NP0	Series input cap
L2	15nH, $\pm 2\%$	0402	—	Shunt inductor
C14	6.8pF, 50V	0402	C0G/NP0	Shunt cap (with L2 forms resonator)
L3	15nH, $\pm 2\%$	0402	—	Series coupling inductor
C15	6.8pF, 50V	0402	C0G/NP0	Center shunt cap
L4	15nH, $\pm 2\%$	0402	—	Shunt inductor
C16	10pF, 50V	0402	C0G/NP0	Series output cap

Note (Place > Note, next to BPF):

437 MHz BPF: 3-pole LC, ~5 MHz bandwidth, target <3 dB insertion loss. ALL capacitors MUST be C0G/NP0 dielectric — X7R has voltage- dependent capacitance that detunes the

filter and generates intermodulation products at RF. Starting values — tune with VNA during bring-up. Required rejection: >30 dB at 145.67 MHz (fundamental), >30 dB at 291.3 MHz (2nd harmonic), >20 dB at 582.7 MHz (4th harmonic).

Section D: MMIC Amplifier (Right)

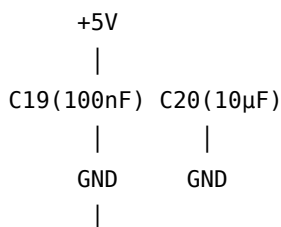
Components

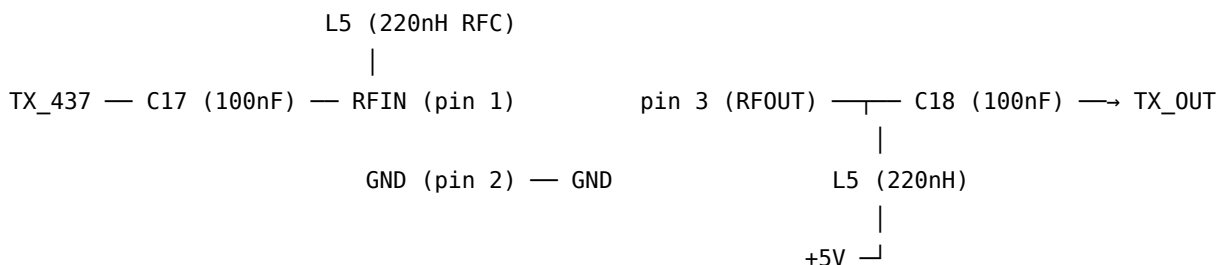
Ref	Value/Part	Package	Connection	Notes
U2	ADL5602ARKZ-R7	SOT-89 (3-pin)	MMIC gain block	20 dB gain, DC-4 GHz
C17	100nF, 16V, X7R	0402	Input DC block	AC coupling in
C18	100nF, 16V, X7R	0402	Output DC block	AC coupling out
L5	220nH, $\pm 5\%$	0805	+5V to RFOUT pin (bias tee)	RFC for DC feed
C19	100nF, 16V, X7R	0402	+5V to GND at L5 junction	Supply bypass
C20	10 μ F, 10V, X5R	0805	+5V to GND	Bulk supply bypass

ADL5602 Pin Connections The ADL5602 is a **3-pin** SOT-89 device. There is no dedicated VCC pin — DC bias enters through the **RFOUT pin** via an external bias tee (inductor L5). This is standard for MMIC gain blocks.

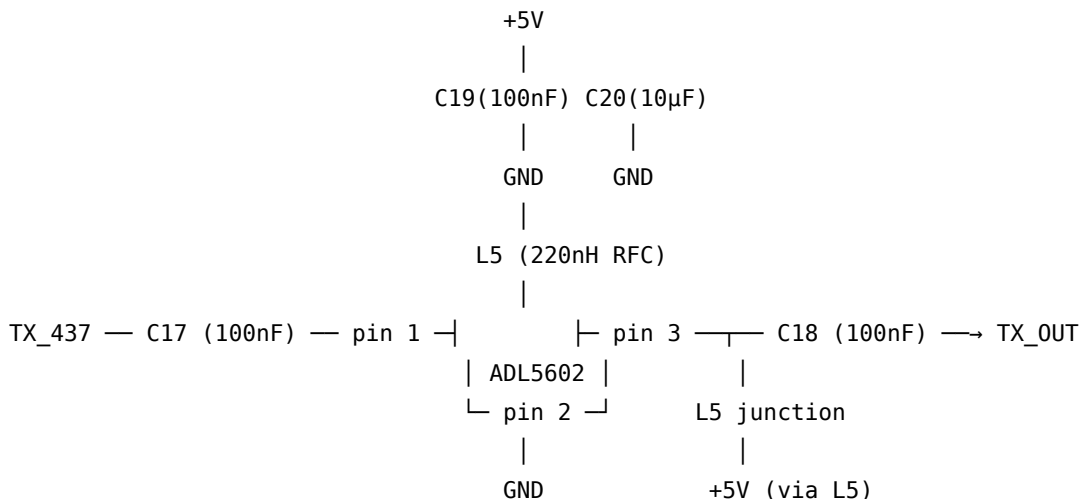
Pin	Name	Connect To	Notes
1	RFIN	TX_437 via C17 (DC block)	RF input from BPF
2	GND	GND	Ground, solder pad to ground plane
3	RFOUT	Bias tee junction (see below)	RF output + DC bias input

Bias tee at RFOUT (pin 3):





Corrected diagram:



The key point: pin 3 (RFOUT) carries both the RF signal out AND the DC supply current in. L5 feeds DC from +5V to pin 3 while blocking RF from reaching the supply. C18 passes RF to the output while blocking the DC bias from reaching the TX SMA.

Note (Place > Note, next to MMIC):

ADL5602: 50Ω matched input/output, 20 dB gain, DC to 4 GHz. 3-pin SOT-89 — NO separate VCC pin. DC bias enters through the RFOUT pin (pin 3) via bias tee inductor L5. This is standard for MMIC gain blocks. L5 (RFC) feeds DC while presenting high impedance at RF. C18 passes RF while blocking DC. Supply current ~60mA from +5V rail. C17 blocks DC at the input. Expected output: -7 dBm (tripler) + 20 dB (MMIC) ≈ +13 dBm at 437 MHz. May need an attenuator pad if output exceeds amateur power limits or if the MMIC saturates — check P1dB in datasheet.

Section E: Output Band-Pass Filter (Far Right)

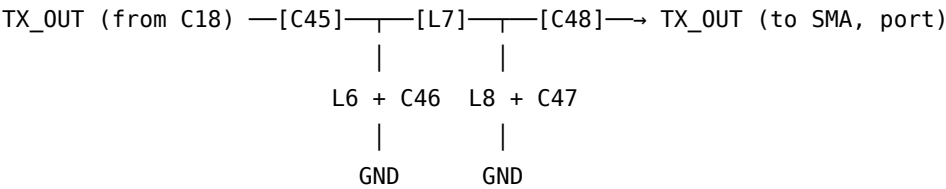
Second 437 MHz BPF, placed after the MMIC. The MMIC has broadband gain (DC–4 GHz) so it amplifies any residual harmonics and spurs that made it through the first BPF, and can generate its own intermodulation products at high output levels. This output filter is the last stage before the

antenna — it ensures clean radiated spectrum and compliance with amateur radio spurious emission limits.

Components

Ref	Value	Package	Dielectric	Notes
C45	10pF, 50V	0402	C0G/NP0	Series input cap
L6	15nH, $\pm 2\%$	0402	—	Shunt inductor
C46	6.8pF, 50V	0402	C0G/NP0	Shunt cap (with L6 forms resonator)
L7	15nH, $\pm 2\%$	0402	—	Series coupling inductor
C47	6.8pF, 50V	0402	C0G/NP0	Center shunt cap
L8	15nH, $\pm 2\%$	0402	—	Shunt inductor
C48	10pF, 50V	0402	C0G/NP0	Series output cap

Same topology and starting values as the pre-MMIC BPF (Section C).

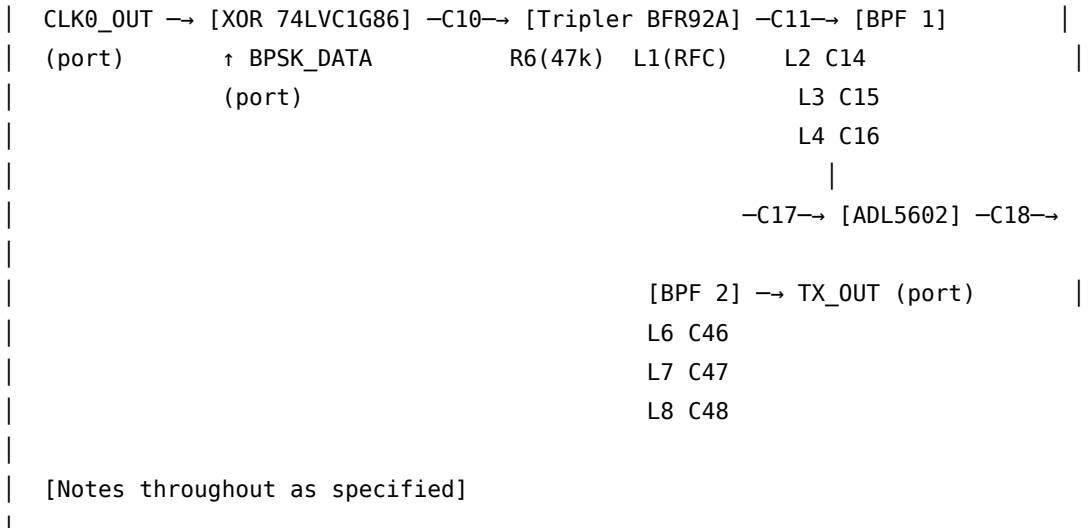


Note (Place > Note, next to output BPF):

Output BPF: identical topology to the pre-MMIC filter. Required because the ADL5602 has broadband gain (DC–4 GHz) and will amplify any residual spurs from the tripler, plus generate its own harmonics at +13 dBm output. FCC Part 97 / ITU amateur satellite rules require spurious emissions at least 43 dB below carrier for transmitters >5W, and good practice for QRP is >30 dB. Two cascaded 3-pole BPFs (pre- and post-MMIC) provide >60 dB total harmonic rejection. Same C0G/NP0 cap and tight-tolerance inductor requirements as the first BPF. Values may need independent tuning from the first BPF since the source/load impedances differ slightly.

Sheet 3 Schematic Layout Suggestion

[Text Frame: TX Chain Description]



Sheet 3 Ports Checklist

- ☐ CLK0_OUT — input, 145.67 MHz from Clock Gen sheet
- ☐ BPSK_DATA — input, from Digital Control (RP2040 GPIO)
- ☐ TX_OUT — output, to Connectors sheet (TX SMA)
- ☐ +3V3 — power port (XOR gate supply)
- ☐ +5V — power port (tripler and MMIC supply)
- ☐ GND — power port (global)

Sheet 4: RX Chain

This sheet implements the 145.9 MHz AFSK uplink receiver: input BPF, mixer, baseband Sallen-Key low-pass filter, and gain stage.

Text Frame (Place > Text Frame, top of sheet):

RX CHAIN — 145.9 MHz FM AFSK uplink receiver. A 2m band-pass filter at the input rejects out-of-band signals before the LNA. A PSA4-5043+ MMIC LNA (~21 dB gain, ~1 dB NF) sets the system noise figure and recovers the conversion loss of the downstream passive mixer. The ADE-1+ double-balanced diode mixer downconverts the 2m signal to baseband using the Si5351A CLK1 as LO; an LC low-pass on the LO line suppresses square-wave harmonics that would otherwise create spurious responses. A 51 Ω resistor terminates the mixer IF port. A 2nd-order Sallen-Key LPF ($f_c \approx 3.3$ kHz) filters the baseband, passing the 1200/2200 Hz AFSK tones while rejecting mixer products and noise. The MCP6022 U7B stage provides ~20 dB gain before the RP2040 ADC. Single-supply operation with 1.65V virtual ground (mid-supply bias). See rx_mixer_trade_study.md for the SA612 → ADE-1+ swap rationale.

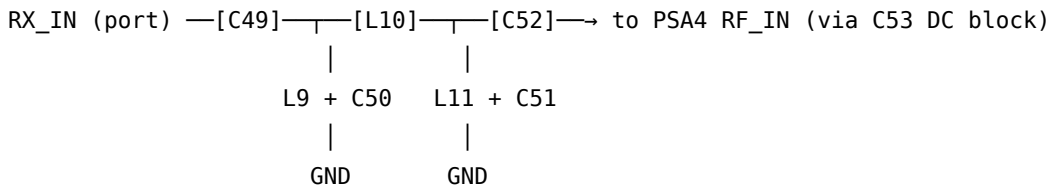
Section A: 2m Input Band-Pass Filter (Far Left)

This filter sits between the RX SMA connector and the PSA4-5043+ LNA input. It passes the 2m amateur band (~144–148 MHz) and rejects out-of-band signals — especially the 437 MHz TX signal which could leak back through the board and saturate the LNA or compress the mixer, and commercial FM broadcast (88–108 MHz) which is extremely strong compared to a satellite uplink.

Components

Ref	Value	Package	Dielectric	Notes
C49	10pF, 50V	0402	C0G/NP0	Series input cap
L9	82nH, $\pm 2\%$	0402	—	Shunt inductor
C50	10pF, 50V	0402	C0G/NP0	Shunt cap (with L9 forms resonator)
L10	82nH, $\pm 2\%$	0402	—	Series coupling inductor
C51	10pF, 50V	0402	C0G/NP0	Center shunt cap
L11	82nH, $\pm 2\%$	0402	—	Shunt inductor
C52	10pF, 50V	0402	C0G/NP0	Series output cap

Starting values — 3-pole LC BPF centered at ~146 MHz. These will need tuning with a VNA during bring-up.



Note (Place > Note, next to 2m BPF):

2m input BPF: 3-pole LC, centered ~146 MHz, ~4 MHz bandwidth. Critical for protecting the LNA front-end and mixer from overload. Must reject: - 437 MHz TX leakage (>40 dB — this is your own transmitter) - FM broadcast 88–108 MHz (>30 dB — these signals are very strong) - 2nd harmonic of LO at 291.8 MHz (>20 dB) ALL capacitors must be C0G/NP0. Starting values are estimates — tune L and C values with VNA to center the passband on 145.9 MHz and achieve <3 dB insertion loss at the operating frequency. If insertion loss is too high, consider a 2-pole filter instead (remove the center L10 coupling and one shunt section).

Section B: LNA → LO LPF → ADE-1+ Mixer → IF Termination (Center-Left)

This section replaces the original SA612 active mixer (EOL part) with a **PSA4-5043+ LNA** in front of an **ADE-1+ passive diode mixer**, plus a 5-pole LC LPF on the LO line to suppress Si5351A square-wave harmonics, and a 51 Ω resistor terminating the IF port. See `rx_mixer_trade_study.md` for the full decision rationale.

The four sub-blocks are presented in signal-flow order: LNA, LO LPF, mixer, IF termination.

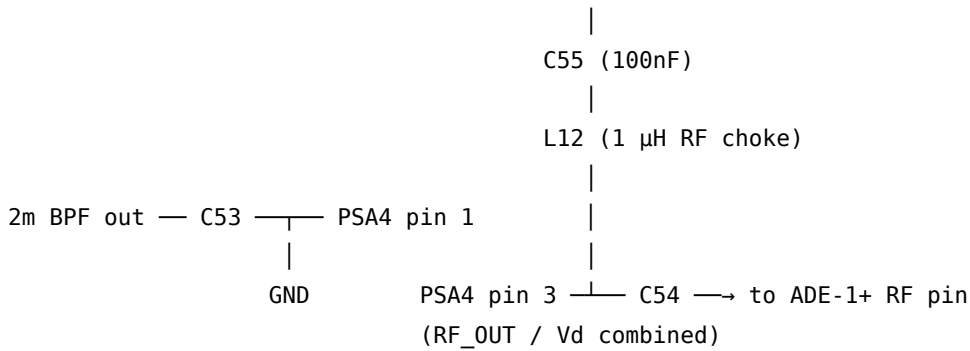
B.1 — PSA4-5043+ LNA

Ref	Value/Part	Package	Connection	Notes
U10	PSA4-5043+	SOT-89 (3 pin + tab)	VHF/UHF LNA	Mini-Circuits, JLC C5240848
C53	100nF, 16V, X7R	0402	Input DC block	2m BPF out → LNA RF_IN
C54	100nF, 16V, X7R	0402	Output DC block	LNA RF_OUT → LO of mixer path
L12	1 μ H, wirewound, SRF > 500 MHz	0805 or 1008	Vd bias choke	Coilcraft 0805CS-102 or equiv RF wirewound
C55	100nF, 16V, X7R	0402	Vd HF bypass	Right at choke top
C56	10 μ F, 10V, X5R	0805	Vd bulk bypass	Within ~10 mm of L12

PSA4-5043+ pin assignment (verify against Mini-Circuits datasheet — SOT-89 case style “WW107”):

Pin	Name	Connect To
1	RF_IN	C53 (DC block) ← 2m BPF output
2	GND	GND (tab also to GND)
3	RF_OUT / Vd	Bias choke L12 to +5V, AND C54 (DC block) → mixer RF input

+5V
|
C56 (10 μ F)

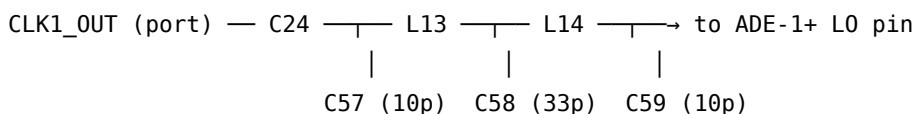


PSA4 pin 2 — GND (also tab to GND)

PSA4-5043+ Note: Self-biased pHEMT LNA. Vd applied at the RF_OUT pin through an RF choke; the DC block C54 separates RF from the bias. Typical performance at 145 MHz: ~21 dB gain, ~1.0 dB NF, +20 dBm P1dB(out), 60 mA at Vd=+5V. The 1 μH wirewound MUST have SRF above 500 MHz — a power inductor with low SRF will resonate inside our band and kill gain. Coilcraft 0805CS series or Murata LQW2BAS family in 1 μH are appropriate. Grounding: tie pin 2 *and* the SOT-89 tab to a solid ground plane with multiple vias. Layout: keep input and output traces short and on opposite sides of the package to avoid feedback.

B.2 — Si5351A LO Low-Pass Filter A 5-pole Butterworth LC LPF, $f_c \approx 200$ MHz, in 50 Ω. The Si5351A output is a CMOS square wave; without filtering, its 3rd harmonic at 437.7 MHz would mix with our own 437 MHz TX leakage and create a spurious response right on top of legitimate RX signals.

Ref	Value	Package	Dielectric	Position
C24	100nF, 16V, X7R	0402	—	DC block on CLK1_OUT (preserved from old design)
C57	10pF, 50V	0402	C0G/NP0	1st shunt cap
L13	68nH, $\pm 2\%$	0402	—	1st series inductor
C58	33pF, 50V	0402	C0G/NP0	Center shunt cap
L14	68nH, $\pm 2\%$	0402	—	2nd series inductor
C59	10pF, 50V	0402	C0G/NP0	Output shunt cap





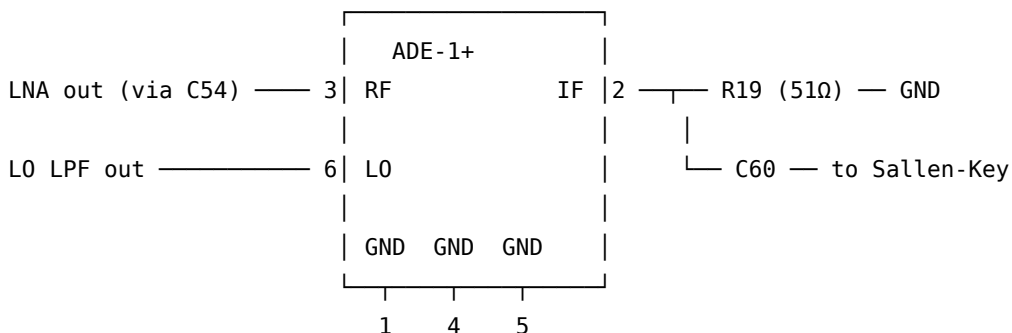
LO LPF Note: 5-pole Butterworth, $f_c \approx 200$ MHz, calculated values rounded to nearest standard. Insertion loss at 145.9 MHz: ~ 0.2 dB. Attenuation at 3rd harmonic (437.7 MHz): ~ 34 dB. Attenuation at 5th harmonic (729.5 MHz): ~ 50 dB. C0G/NP0 dielectric is mandatory on all three caps — X7R will detune the filter. Inductors should be $\pm 2\%$ tolerance or better; 0402 multilayer chip inductors with $Q > 25$ at 150 MHz are appropriate (e.g., Murata LQG15HS or Coilcraft 0402CS). Verify on VNA before fab — exact LPF response is sensitive to PCB parasitics at these frequencies.

B.3 — ADE-1+ Mixer

Ref	Value/Part	Package	Notes
U3	ADE-1+	CD636 (7.87×5.59 mm, 6-pad SMT)	Mini-Circuits passive diode ring, Level 7

ADE-1+ pin connections (per datasheet C2942210.pdf):

Pin	Name	Connect To
1	GND	GND
2	IF	R19 ($51\ \Omega$ term) to GND, then DC block C60 to Sallen-Key input
3	RF	C54 (LNA output DC block) $\leftarrow$ PSA4
4	GND	GND
5	GND	GND
6	LO	LO LPF output $\leftarrow$ Si5351A CLK1 chain



| | |
GND GND GND

ADE-1+ Note: Passive double-balanced diode-ring mixer, 0.5–500 MHz LO/RF, DC–500 MHz IF. **No DC supply required** — transformer-coupled on all three ports. Typical performance at our band: ~5 dB conversion loss, ~55 dB LO-RF isolation, ~45 dB LO-IF isolation, +15 dBm IIP3, +1 dBm P1dB. LO drive: +7 dBm specified, +4 dBm to +10 dBm tolerable with mild conversion-loss penalty. Max RF power 50 mW (+17 dBm). Package footprint (CD636) must be drawn in Altium — see recommended PCB land pattern in datasheet. Tie all three GND pads (1, 4, 5) to ground plane with vias underneath, kept short.

B.4 — IF Port Termination

Ref	Value	Package	Connection
R19	51 Ω , 1%	0402	ADE-1+ pin 2 (IF) to GND
C60	100nF, 16V, X7R	0402	ADE-1+ pin 2 → Sallen-Key R8 (DC block)

IF Termination Note: Passive diode mixers require the IF port to see ~50 Ω across the band of interest, otherwise the diode bridge doesn't balance properly and IP3/conversion loss degrade sharply. The Sallen-Key input (22 k Ω at R8) is essentially open-circuit to the mixer — so we add R19 (51 Ω) directly across the IF port to GND. The Sallen-Key just senses the voltage across this resistor through DC block C60. For more sophisticated designs, replace R19 with a diplexer (RC LPF + RL HPF) to maintain 50 Ω at *all* frequencies, but a single resistor is adequate for AFSK at baseband.

Section C: Sallen-Key Low-Pass Filter (Center)

2nd-order Butterworth LPF, $f_c \approx 3.3$ kHz. Passes AFSK tones (1200/2200 Hz) and rejects mixer products and wideband noise.

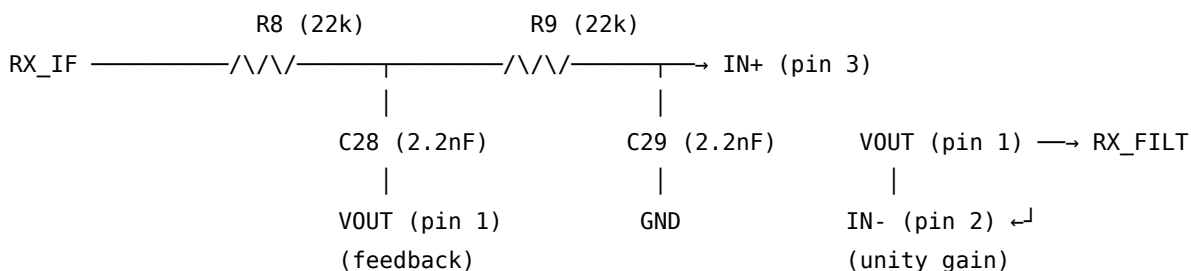
Components

Ref	Value	Package	Dielectric	Connection
R8	22k Ω , 1%	0402	—	Series input to junction

Ref	Value	Package	Dielectric	Connection
R9	22k Ω , 1%	0402	—	Junction to op-amp IN+
C28	2.2nF, 25V, C0G/NP0	0402	—	Junction (R8-R9) to op-amp output (feedback)
C29	2.2nF, 25V, C0G/NP0	0402	—	Op-amp IN+ to GND

Op-amp: MCP6022 U7A (half of dual MCP6022, pins 1-3)

MCP6022 Pin	Name	Connect To
1	VOUT_A	Output → feedback to C28, → C30 (AC couple to gain stage)
2	VIN-_A	Direct wire to VOUT_A (pin 1) — unity gain feedback
3	VIN+_A	From R9, C29 to GND, C28 from junction
8	VDD	+3V3 via C31 (100nF bypass)
4	VSS	GND



Note (Place > Note, next to Sallen-Key):

Sallen-Key topology per Art of Electronics Fig 6.16A. C28 is the FEEDBACK capacitor — connects from the R8/R9 junction to the op-amp output. C29 goes from IN+ to GND. This is NOT two caps to ground — C28's feedback to the output is what creates the complex pole pair and gives -40 dB/decade rolloff (2nd order). With R8=R9 and C28=C29 (equal-component Butterworth), $Q = 0.707$, no ringing. $f_c = 1/(2\pi \times R \times C) = 1/(2\pi \times 22k \times 2.2nF) = 3,289$ Hz. LTspice verified: -3dB at 2,985 Hz, 1200 Hz passes at -0.56 dB, 2200 Hz at -1.75 dB. C0G/NP0 caps required for filter accuracy.

Section D: Baseband Gain Stage (Right)

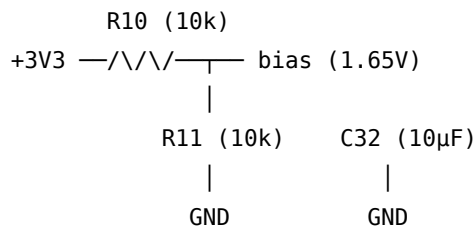
MCP6022 U7B provides voltage gain to bring the baseband signal to a level suitable for the RP2040's 12-bit ADC (0–3.3V range).

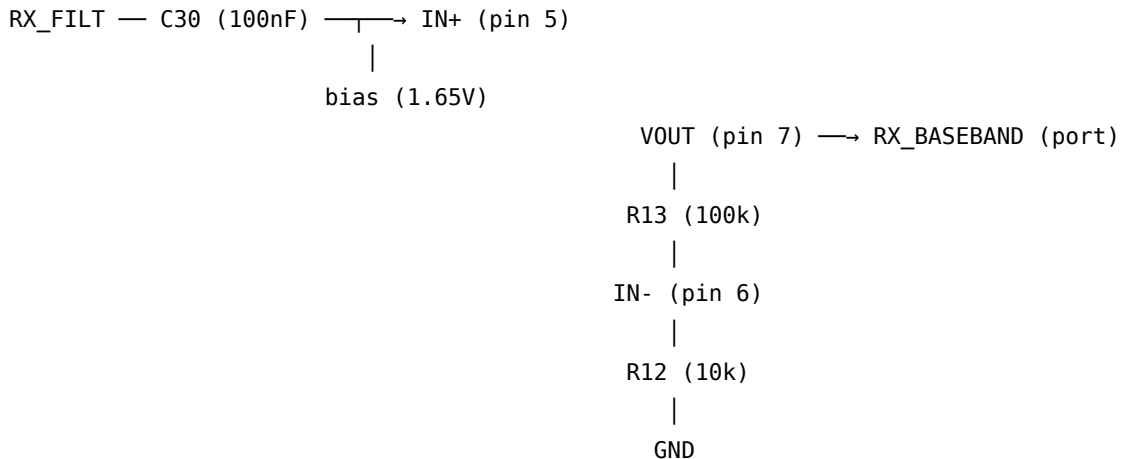
Components

Ref	Value	Package	Connection	Notes
C30	100nF, 16V, X7R	0402	Sallen-Key output to gain stage input	AC coupling (removes DC offset)
R10	10k Ω , 1%	0402	+3V3 to bias junction	Bias divider top
R11	10k Ω , 1%	0402	Bias junction to GND	Bias divider bottom
C32	10 μ F, 10V, X5R	0805	Bias junction to GND	Bias decoupling (AC ground)
R12	10k Ω , 1%	0402	Non-inverting amp Rg (IN- to GND)	Gain-setting ground resistor
R13	100k Ω , 1%	0402	Feedback (VOUT to IN-)	Gain = $1 + R_f/R_g = 11$ (20.8 dB)
C31	100nF, 16V, X7R	0402	MCP6022 VDD (pin 8) to GND	Supply bypass

Op-amp: MCP6022 U7B (second half, pins 5-7)

MCP6022 Pin	Name	Connect To
7	VOUT_B	Output → R13 feedback, → RX_BASEBAND port
6	VIN-_B	R12 to GND, R13 from VOUT_B
5	VIN+_B	C30 input, R10/R11 bias junction
8	VDD	+3V3 (shared with U7A, one bypass cap C31)
4	VSS	GND (shared with U7A)

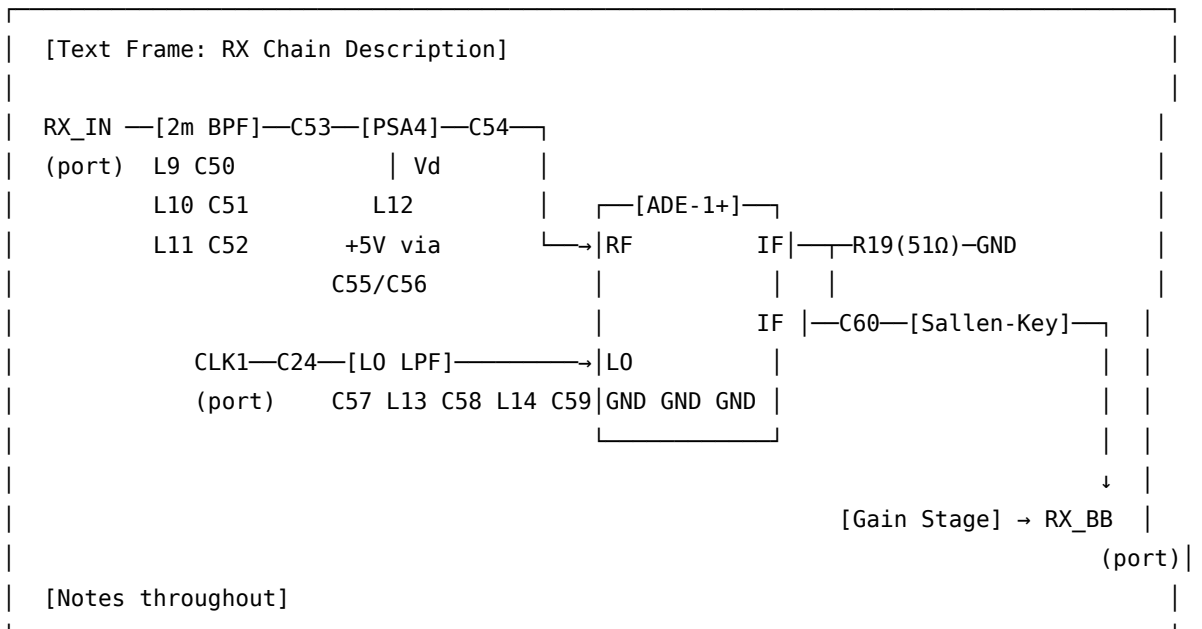




Note (Place > Note, next to gain stage):

Non-inverting amplifier, gain = $1 + 100k/10k = 11$ (20.8 dB). R10/R11 bias the input to 1.65V (mid-supply) for single-supply operation. C32 (10 μ F) decouples the bias point at AC so it acts as a solid virtual ground for the signal. C30 AC-couples the Sallen-Key output to remove any DC offset from the filter. Output goes to RP2040 ADC (GP26/ADC0). Full-scale ADC input is 0–3.3V; with 1.65V bias, the signal can swing $\pm 1.65V$ before clipping. Gain of 11 may need adjustment based on actual signal levels — change R13 to tune. MCP6022 GBW = 10 MHz, well above the ~3 kHz signal band.

Sheet 4 Schematic Layout Suggestion



Sheet 4 Ports Checklist

- ☐ CLK1_OUT — input, 145.9 MHz LO from Clock Gen sheet (drives LO LPF → ADE-1+)
- ☐ RX_IN — input, from Connectors sheet (RX SMA)
- ☐ RX_BASEBAND — output, to Digital Control (Pico ADC)
- ☐ +3V3 — power port (MCP6022 supply)
- ☐ +5V — power port (PSA4-5043+ LNA Vd via L12 choke)
- ☐ GND — power port (global)

Sheet 5: Digital Control

This sheet holds the RP2040 (Pico module), USB connector, and GPIO assignments.

Text Frame (Place > Text Frame, top of sheet):

DIGITAL CONTROL — Raspberry Pi Pico module (RP2040-based). Handles BPSK/DBPSK encoding for TX, AFSK demodulation for RX, Si5351A configuration via I2C, and command processing. Using Pico module for prototype — all RP2040 support circuitry (flash, regulator, crystal) is on-module. Bare RP2040 redesign deferred to v2 flight board after RF chain validation.

Section A: Pico Module (Center of Sheet)

Place a 2×20 pin header symbol (2.54mm pitch) or a Pico module symbol if available in Manufacturer Part Search. Label each pin with its net name.

Pico Module Pin Connections

Pico Pin	Net Name	Goes To	Connection
GP2 (pin 4)	BPSK_DATA	XOR gate pin 2	BPSK_DATA net label
GP3 (pin 5)	COMMS_IRQ	CSKB H1.16	COMMS_IRQ net label (output, active-low, push-pull, normally high — asserts when RX packet ready or TX status change; IHU ISR triggers SPI read)
GP4 (pin 6)	I2C_SDA	Si5351A SDA	I2C_SDA net label
GP5 (pin 7)	I2C_SCL	Si5351A SCL	I2C_SCL net label
GP16 (pin 21)	SPI_COMMS_MISO	CSKB H1.22	SPI0 MISO — comms → IHU (slave TX)
GP17 (pin 22)	SPI_COMMS_CS_N	CSKB H1.24	SPI0 CSn — slave select from IHU

Pico Pin	Net Name	Goes To	Connection
GP18 (pin 24)	SPI_COMMS_SCK	CSKB H1.21	SPI0 SCK — clock from IHU
GP19 (pin 25)	SPI_COMMS_MOSI	CSKB H1.23	SPI0 MOSI — IHU → comms
GP26/ADC0 (pin 31)	RX_BASEBAND	MCP6022 output	RX_BASEBAND net label
3V3 (pin 36)	+3V3	Power (from EPS)	+3V3 power port
GND (pins 3,8,13,18,23,28,33,38)	GND	Ground	GND power port
VSYS (pin 39)	VSYS	System power input	See Power section
VBUS (pin 40)	VBUS_USB	From USB connector VBUS	USB power

Note (Place > Note, next to Pico module):

Pico module power: The Pico has an internal regulator that produces 3.3V from VSYS (1.8–5.5V input). For stack operation, feed +5V from the CSKB bus to VSYS through a Schottky diode (prevents back-feeding when USB is also connected). The Pico’s internal 3V3 regulator is then active and supplies the on-module 3V3. The +3V3 power port on this board is the EPS-supplied rail — connect Pico pin 36 (3V3 OUT) to +3V3 only if you want the Pico to be powered from the bus. Do NOT connect both VSYS and 3V3(OUT) to the bus simultaneously without understanding the power path.

Section B: USB Connector (Left of Pico)

For prototype, the Pico module’s own Micro-USB connector is the primary programming and debug interface. A separate USB connector on the comms board is **optional** — skip for prototype and use the Pico’s built-in USB.

If adding a separate connector:

Ref	Part	Package	Connection
J1	USB Micro-B or USB-C	SMD	D+ to Pico USB_DP, D- to USB_DN

Note (Place > Note):

USB connector is optional for prototype. The Pico module has its own Micro-USB. For the flight board (v2 with bare RP2040), a USB-C connector would be placed here with 27Ω termination resistors on D+/D-.

Section C: Debug/Programming Header (Optional, Bottom-Left)

Ref	Part	Pins	Connection
J2	1×4 pin header, 2.54mm	4	SWCLK, SWDIO, GND, 3V3

Connect to Pico SWD pins: SWCLK (Pico debug pin), SWDIO (Pico debug pin), GND, 3V3. Useful for debugging with a probe but not required if using USB serial for bring-up.

Section D: Status LEDs (Optional, Bottom-Right)

Ref	Color	Connection	Indicates
D1	Green	GPIO → R14 (330Ω) → LED → GND	TX active
D2	Yellow	GPIO → R15 (330Ω) → LED → GND	RX active
D3	Green	+3V3 → R16 (1kΩ) → LED → GND	Board power

Use whatever free GPIOs are available. Mark as DNP for flight.

Section E: Spare GPIO Header (Recommended)

Ref	Part	Pins	Connection
J3	1×6 pin header, 2.54mm	6	Unused GPIOs (GP10–GP15 or similar)

Note (Place > Note, next to spare header):

Spare GPIO header: invaluable during integration and testing. Can be used for UART debug output, logic analyzer probing, or connecting additional sensors during orbit. Label pins with GPIO numbers.

Sheet 5 Schematic Layout Suggestion

[Text Frame: Digital Control Description]

[USB Connector]	[Pico Module / 2x20 Header]
(optional)	GP2 — BPSK_DATA (port)
D+ D- VBUS GND	GP3 — COMMS_IRQ (port → H1.16)
	GP4 — I2C_SDA (port)
	GP5 — I2C_SCL (port)
	GP16 — SPI_COMMS_MISO (port → H1.22)
	GP17 — SPI_COMMS_CS_N (port → H1.24)
	GP18 — SPI_COMMS_SCK (port → H1.21)
	GP19 — SPI_COMMS_MOSI (port → H1.23)
	GP26 — RX_BASEBAND (port)
	3V3 — +3V3 (power port)
	VSYS — from +5V via Schottky
	GND — GND (power port)
[Debug Header]	[Spare GPIO Header]
SWCLK SWDIO GND	GP10 GP11 GP12 GP13 GP14 GP15
[Status LEDs]	
D1(TX) D2(RX) D3(PWR)	

Sheet 5 Ports Checklist

- ☐ I2C_SDA — bidirectional, to GP4
- ☐ I2C_SCL — bidirectional, to GP5
- ☐ BPSK_DATA — output, from GP2
- ☐ COMMS_IRQ — output, from GP3 (to Connectors sheet → H1.16)
- ☐ SPI_COMMS_MISO — output, from GP16 (to Connectors sheet → H1.22)
- ☐ SPI_COMMS_CS_N — input, to GP17 (from Connectors sheet → H1.24)
- ☐ SPI_COMMS_SCK — input, to GP18 (from Connectors sheet → H1.21)
- ☐ SPI_COMMS_MOSI — input, to GP19 (from Connectors sheet → H1.23)
- ☐ RX_BASEBAND — input, to GP26/ADC0
- ☐ +3V3 — power port (Pico 3V3)
- ☐ GND — power port (global)

Note (Place > Note, next to GP3 / COMMS_IRQ label):

COMMS_IRQ is a data-ready interrupt from the comms RP2040 to the internal housekeeping unit, modelled after the AX5043 IRQ pattern in AMSAT RT-IHU. Push-pull output, active-low, normally held high. Comms firmware asserts the line when (a) an RX packet has been fully received and is available in the slave FIFO, or (b) a TX-complete or TX-underrun status needs IHU attention. The IHU uses a GPIO-EXTI ISR to trigger a SPI read

transaction; comms de-asserts the line after the IHU CS_N drops (or after a firmware-side clear). Do NOT add a pull-up on this board — the IHU side (R11 on IHU Sheet 4) has a 10k pull-up that keeps the line defined while comms is in reset.

Sheet 6: Power

This sheet handles power distribution across the comms board. **All regulated power (+3V3, +5V) comes from the EPS board via the CSKB stack connector** — there are no on-board voltage regulators on the comms board (prototype configuration).

Text Frame (Place > Text Frame, top of sheet):

POWER DISTRIBUTION — The comms board receives +3V3 and +5V from the EPS via the CSKB stack connectors. No on-board voltage regulation. This sheet provides input protection, bulk bypass capacitors, and distributes power to all subsystems. Power budget: ~150 mA on +3V3 (RP2040 + Si5351A + logic), ~140 mA on +5V (PSA4-5043+ LNA + ADL5602 MMIC + tripler bias). Total: <300 mA — well within EPS TPS62933F capacity (3A per rail).

Section A: Input Protection and Filtering (Left of Sheet)

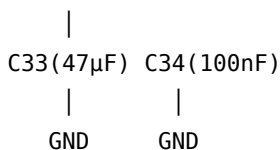
Power enters the board through the CSKB H2 connector (defined on the Connectors sheet). On this sheet, we receive the power port symbols and add protection and filtering.

Components

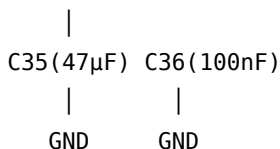
Ref	Value	Package	Connection	Purpose
D4	Schottky (SS14 or BAT54S)	SMA or SOT-23	+5V line, anode to bus, cathode to internal +5V	Reverse polarity protection
C33	47μF, ≥10V (Panasonic EEE-FK1V470P wet Al-electrolytic or EEH-ZA1V470P polymer)	SMD can (~6.3mm)	+5V (after protection) to GND	Input bulk capacitor
C34	100nF, 16V, X7R	0402	+5V to GND	HF bypass at entry

Ref	Value	Package	Connection	Purpose
C35	47 μ F, $\geq$ 10V (Panasonic EEE-FK1V470P wet Al-electrolytic or EEH-ZA1V470P polymer)	SMD can (~6.3mm)	+3V3 to GND	Input bulk capacitor
C36	100nF, 16V, X7R	0402	+3V3 to GND	HF bypass at entry

+5V (from CSKB H2.25/26) — D4 (Schottky) — +5V rail (internal)



+3V3 (from CSKB H2.27/28) — +3V3 rail



Note (Place > Note, next to input protection):

D4 provides reverse polarity protection on the +5V rail. Adds ~0.3V forward drop — the ADL5602 and PSA4-5043+ LNA can tolerate this (ADL5602 specified for supply down to 4.5V; PSA4 Vd range is +3V to +5V, so 4.7V is within window). At 4.7V the PSA4 Id is slightly higher than at 5.0V — check current budget headroom. The +3V3 rail does NOT have a series diode because the RP2040 and Si5351A are sensitive to voltage drop — 3.0V may be too low for reliable operation. If reverse polarity protection is needed on +3V3, use a P-MOSFET instead of a Schottky (near-zero drop when conducting). Bulk caps dampen transients from the CSKB connector traces. The EPS has its own bulk caps at the connector — these are the receiving-end complements.

Section B: Per-Subsystem Bypass (Center of Sheet)

Each IC has local bypass caps on its own sheet (Clock Gen, TX, RX, Digital Control). This section provides additional board-level bulk bypass at the power distribution point.

Components

Ref	Value	Package	Connection	Purpose
C37	10 μ F, 10V, X5R	0805	+3V3 to GND	Board-level bulk, 3.3V
C38	100nF, 16V, X7R	0402	+3V3 to GND	Board-level HF, 3.3V
C39	10 μ F, 10V, X5R	0805	+5V to GND	Board-level bulk, 5V
C40	100nF, 16V, X7R	0402	+5V to GND	Board-level HF, 5V

Note (Place > Note, next to bulk bypass):

Board-level bulk bypass at the power distribution star point. Local bypass at each consumer IC is the responsibility of that IC's sheet (C1/C2/C3 on Clock Gen, C8 on TX, C21/C22 on RX, etc.). These caps cover trace inductance between the CSKB connector and the far side of the board. Place C37/C39 near the center of their respective power distribution traces.

Section C: Pico Module Power Feed (Right of Sheet)

The Pico module needs power on its VSYS pin. Feed it from the +5V bus through a Schottky diode so the Pico's internal regulator handles its own 3.3V generation. The diode prevents back-feeding from USB when the Pico is also connected to a computer.

Components

Ref	Value	Package	Connection	Purpose
D5	Schottky (BAT54 or SS14)	SOT-23 or SMA	+5V to Pico VSYS	Isolation diode

+5V rail — D5 (Schottky) → VSYS (to Pico pin 39)

Note (Place > Note, next to D5):

D5 isolates the stack +5V from the Pico's USB VBUS. When USB is connected, the Pico's internal diode on VBUS feeds VSYS. When running from the stack (no USB), D5 feeds VSYS from +5V. The Pico's internal 3.3V buck-boost regulator produces 3V3 from VSYS. Drop across D5 (~0.3V) gives VSYS $\approx$ 4.7V — well within the Pico's VSYS range (1.8–5.5V).

Section D: Power Indicator LEDs (Optional, Bottom)

Ref	Color	Connection	Indicates
D6	Green	+3V3 → R17 (1k Ω) → LED → GND	3.3V rail active

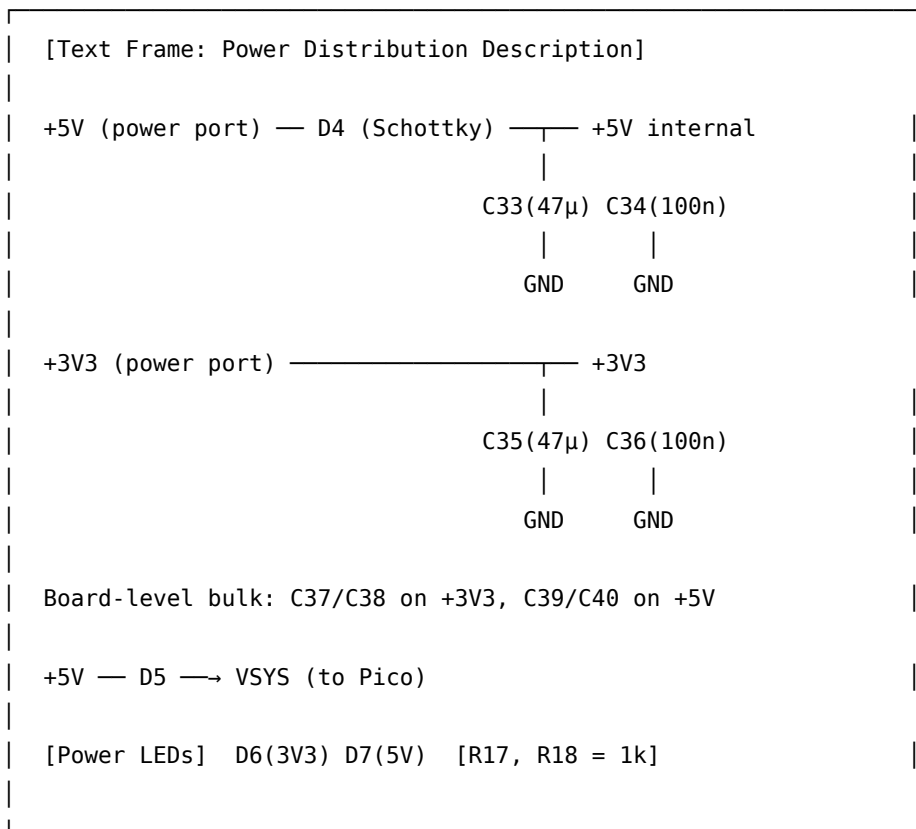
Ref	Color	Connection	Indicates
D7	Green	+5V → R18 (1kΩ) → LED → GND	5.0V rail active

Mark as DNP (Do Not Populate) for flight — LEDs waste power in orbit.

Note (Place > Note, next to LEDs):

Power indicator LEDs for bench debugging. $I = (3.3-2.0)/1k = 1.3mA$ (dim but visible).
Mark R17/R18 as DNP-optional in BOM for flight.

Sheet 6 Schematic Layout Suggestion



Sheet 6 Ports Checklist

- ☐ +3V3 — power port (global, from EPS via CSKB H2.27/28)
- ☐ +5V — power port (global, from EPS via CSKB H2.25/26)
- ☐ GND — power port (global)
- ☐ VSYS — net label, to Pico module VSYS pin

Important Note on AMS1117-3.3 and MCP1701T-5002I

The v0.1 prototype schematic includes U5 (MCP1701T-5002I/MB, 5V LDO) and U6 (AMS1117-3.3, 3.3V LDO) on the Power sheet. **These are no longer needed** since power comes from the EPS via the bus connector.

Action: Remove U5, U6, and their associated input/output caps from the Power sheet. If you want to keep them as a fallback for standalone bench testing (without the EPS board), move them to a “standalone power” section with a clear note that they are DNP when operating in the stack.

If keeping the AMS1117-3.3 for standalone operation: it **requires** a $\geq 10\mu\text{F}$ output capacitor for stability (the current v0.1 schematic is missing this cap — add it). The MCP1701T-5002I/MB needs minimum 5.35V input ($V_{\text{out}} + \text{dropout} = 5.0 + 0.35 = 5.35\text{V}$), so it needs a bench supply of at least 5.5V.

Sheet 7: Connectors

This sheet holds the CSKB H1 + H2 stack connectors, RF SMA connectors, and any test point headers. It is the physical interface between the comms board and the rest of the CubeSat stack.

Text Frame (Place > Text Frame, top of sheet):

CONNECTORS — CSKB H1 + H2 stack connectors for inter-board power and signal distribution, SMA connectors for TX and RX antenna ports, and test points for bench validation. All power (+3V3, +5V) arrives on H2; I2C, SPI, and COMMS_IRQ route through H1. The comms board is a power CONSUMER — it receives regulated rails from the EPS.

Section A: CSKB H1 + H2 Stack Connectors (Center of Sheet)

Connector Selection Part: 2× Samtec ESQ-126-39-G-D — 2×26 (52-pin) each, 0.1" (2.54 mm) pitch, stackthrough through-hole vertical socket. One populates the H1 (signals) position, one populates the H2 (power) position, for a total of 104 pins per Pumpkin’s CSKB layout. Same stackthrough variant as the IHU board; EPS uses the non-stackthrough ESQ-126-37-G-D because it is the stack endpoint.

The mating adjacent board’s CSKB sockets are the same ESQ-126-39-G-D on IHU and payload, and ESQ-126-37-G-D on EPS. Default stacking height is 15 mm.

Search Manufacturer Part Search for “ESQ-126-39-G-D” or place a generic 2×26 pin socket symbol (0.1" (2.54 mm) pitch) and assign the Samtec MPN in the component properties. **Mechanical note:** exact X/Y position of the H1 and H2 footprints on the PCB must match Pumpkin’s motherboard layout (see DS_CSK_MB_710-00484-E.pdf page 5, “Simplified Mechanical Layout”) before layout.

Pin Assignment (Pumpkin CSKB H1/H2)

Canonical reference: `system/interfaces/cskb_pinmap.md` is the authoritative pin map for the centisat stack. The table below is a board-local view and must match that file. Update the canonical file first if anything here needs to change.

Pin numbers follow the Pumpkin CubeSat Kit Motherboard H1/H2 CSKB convention (DS_CSK_MB_710-00484-E.pdf, doc Rev. A, March 2012, pp. 13–16). The comms board uses only the subset it needs; all other H1/H2 pins are left unconnected on this board but must remain in the bus routing for other subsystems.

Pin	Pumpkin name	centisat net	Direction (Comms)	Function
H1.16	I0.8	COMMS_IRQ	Out	Data-ready interrupt to IHU (active-low, push-pull)
H1.21	I0.3 (SCK0)	SPI_COMMS_SCK	In	SPI clock from IHU
H1.22	I0.2 (SDI0)	SPI_COMMS_MISO	Out	SPI data, comms → IHU
H1.23	I0.1 (SDO0)	SPI_COMMS_MOSI	In	SPI data, IHU → comms
H1.24	I0.0 (-CS_SD)	SPI_COMMS_CS_N	In	SPI chip select from IHU
H1.41	SDA_SYS	I2C_SDA	Bidirectional	Shared housekeeping I2C data
H1.43	SCL_SYS	I2C_SCL	Bidirectional	Shared housekeeping I2C clock
H1.49	USER2	EPS_ALERT_N	(not used on comms)	EPS LTC4162 alert — consumed by IHU only
H2.25	+5V_SYS	+5V	Power In	5 V stack rail (from EPS)
H2.26	+5V_SYS	+5V	Power In	5 V (parallel for current sharing)
H2.27	VCC_SYS	+3V3	Power In	3.3 V stack rail (from EPS)
H2.28	VCC_SYS	+3V3	Power In	3.3 V (parallel for current sharing)

Pin	Pumpkin name	centisat net	Direction (Comms)	Function
H2.29	DGND	GND	Power	Digital ground return
H2.30	DGND	GND	Power	Digital ground return
H2.31	AGND	GND	Power	Analog ground (tied to GND single plane)
H2.32	DGND	GND	Power	Digital ground return (third DGND pin)
H2.45	VBATT	VBAT	Power In (monitor only)	Battery bus for ADC telemetry
H2.46	VBATT	VBAT	Power In (monitor only)	Battery bus (parallel)

All other H1/H2 pins are reserved or left unconnected on the comms board.

Note (Place > Note, next to CSKB connectors):

CSKB stack connectors: 2× Samtec ESQ-126-39-G-D (2×26, 0.1" (2.54 mm), stackthrough headers, one each for H1 and H2). Pin assignment locked to the Pumpkin CSKB H1/H2 convention — all boards in the centisat stack (EPS, IHU, comms, payload) use the same H1/H2 pin numbers for the same signal classes. The comms board is a power CONSUMER on all power pins and a SPI SLAVE to the IHU. COMMS_IRQ (H1.16) is driven by the comms RP2040 as a data-ready interrupt to the IHU. I2C is shared between EPS LTC4162 (0x68) and comms Si5351A (0x60) — no address conflict. See `system/interfaces/cskb_pinmap.md` for the full canonical pin map.

Bulk Bypass at Stack Connector Place bypass caps physically close to the CSKB H2 connector pads:

Ref	Value	Package	Connection	Purpose
C41	10μF, 10V, X5R	0805	+3V3 to GND at connector	Receiving-end bulk, 3.3V
C42	100nF, 16V, X7R	0402	+3V3 to GND at connector	Receiving-end HF, 3.3V
C43	10μF, 10V, X5R	0805	+5V to GND at connector	Receiving-end bulk, 5V
C44	100nF, 16V, X7R	0402	+5V to GND at connector	Receiving-end HF, 5V

Note (Place > Note, next to bypass caps):

Connector-side bypass caps are the receiving complement to the EPS board's connector-side caps (C60–C63 on EPS Sheet 3). Together they form a low-impedance distributed bypass across the stack connector. Place these caps as close to the connector pins as possible — trace inductance between connector and cap defeats the purpose.

Section B: RF Connectors (Top-Right)**Components**

Ref	Part	Connection	Purpose
J4	SMA edge-launch, female	TX_OUT net	437 MHz TX antenna port
J5	SMA edge-launch, female	RX_IN net	145.9 MHz RX antenna port

Search Manufacturer Part Search for “SMA edge launch” or “SMA PCB connector”. For prototype, through-hole SMA connectors work fine. For flight, use edge-launch SMAs with ground tabs soldered to the ground plane.

TX_OUT (port) → J4 (SMA) → to TX antenna / cable

RX_IN (port) → J5 (SMA) → from RX antenna / cable

Note (Place > Note, next to SMA connectors):

Separate TX and RX SMA connectors — no diplexer for prototype. For flight, a diplexer can be added between a single antenna and the two SMA ports, or external to the board. SMA center pin impedance is 50Ω — match traces accordingly during PCB layout. Keep SMA-to-first-component traces as short as possible, with continuous ground plane underneath. Controlled-impedance traces (50Ω microstrip or CPWG) required for RF paths.

Section C: Test Points (Bottom of Sheet)**Test Point List**

Ref	Net	Purpose	Measurement
TP1	+3V3	3.3V rail	Rail voltage, ripple (scope)
TP2	+5V	5.0V rail	Rail voltage, ripple (scope)

Ref	Net	Purpose	Measurement
TP3	CLK0_OUT	Si5351A CLK0	145.67 MHz carrier (spectrum analyzer)
TP4	BPSK_145	XOR output	Modulated 145.67 MHz (scope/SA)
TP5	TRIPLER_OUT	Tripler output	Unfiltered harmonics (SA)
TP6	TX_437	Post-BPF	Filtered 437 MHz (SA)
TP7	RX_IF	ADE-1+ IF output (post-termination, pre-DC-block)	Baseband signal (scope)
TP8	RX_BASEBAND	MCP6022 output	Amplified baseband (scope)
TP9	GND	Ground reference	Scope ground clip point

Place TP9 (GND) as a large pad or loop — always need a convenient ground reference for scope probing. Consider placing two GND test points on opposite sides of the board.

How to Place Test Points in Altium

1. Place > Part > search “Test Point” in Manufacturer Part Search
2. Or: create a generic 1-pin schematic symbol called “TP”
3. Assign a footprint: round pad (1–2mm diameter) or Keystone 5019 loop
4. Label each test point with its net name in the Comment field
5. On the PCB, test points appear as pads — probe with scope/DMM

Note (Place > Note, next to test points):

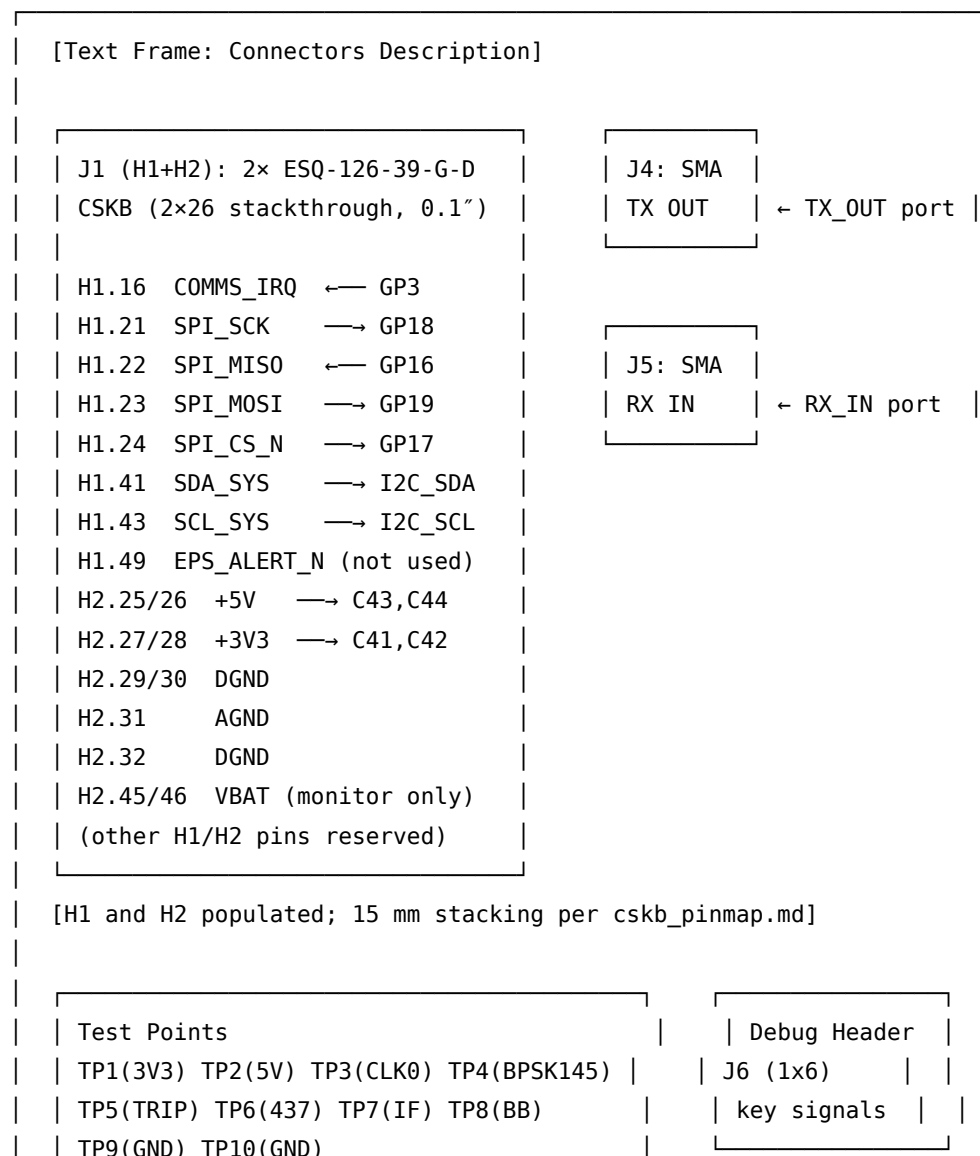
Test points are critical for bring-up and debugging. TP3–TP6 trace the TX signal through the chain — essential for validating each stage independently. TP7–TP8 trace the RX chain. Use short probe leads on RF test points to minimize added inductance. TP9 (GND) should be a loop-style test point for easy scope ground clip attachment. For flight boards, reduce to TP1/TP2 (power) + TP9 (GND) minimum.

Section D: Test/Debug Header (Optional, Bottom-Right)

Ref	Part	Pins	Connection
J6	1×6 pin header, 2.54mm	6	BPSK_DATA, I2C_SCL, I2C_SDA, CLK0_OUT, RX_BASEBAND, GND

This header brings key signals to a convenient probe point for logic analyzer or oscilloscope connection during integration testing.

Sheet 7 Schematic Layout Suggestion



[Notes throughout as specified]

Sheet 7 Ports Checklist

- ☐ TX_OUT — input, from TX Chain (to TX SMA J4)
- ☐ RX_IN — output, from RX SMA J5 (to RX Chain)
- ☐ I2C_SCL — bidirectional, to CSKB bus (H1.43)
- ☐ I2C_SDA — bidirectional, to CSKB bus (H1.41)
- ☐ COMMS_IRQ — output, to CSKB bus (H1.16) from Digital Control GP3
- ☐ SPI_COMMS_SCK — input, from CSKB bus (H1.21) to Digital Control GP18
- ☐ SPI_COMMS_MISO — output, to CSKB bus (H1.22) from Digital Control GP16
- ☐ SPI_COMMS_MOSI — input, from CSKB bus (H1.23) to Digital Control GP19
- ☐ SPI_COMMS_CS_N — input, from CSKB bus (H1.24) to Digital Control GP17
- ☐ EPS_ALERT_N — input (reserved), from CSKB bus (H1.49) — not wired in v0.1
- ☐ +3V3 — power port, from CSKB (H2.27, H2.28)
- ☐ +5V — power port, from CSKB (H2.25, H2.26)
- ☐ GND — power port, from CSKB (H2.29, H2.30, H2.31 AGND, H2.32)
- ☐ VBAT — power port, from CSKB (H2.45, H2.46) — monitor only

After All Seven Sheets Are Done

1. Annotate

Tools > Annotate Schematic Quietly (or Annotate Schematic for manual control). Assign unique designators across all sheets: R1, R2, ... C1, C2, ... U1, U2, ... etc.

2. Compile and Validate

Project > Validate PCB Project. Check the Messages panel for: - **Net mismatch** — net labels with the same name on different sheets must match exactly (case-sensitive) - **Unconnected pins** — add no-connect X markers on any intentionally unconnected pins (e.g., Si5351A CLK2, unused Pico GPIOs) - **Duplicate designators** — fix before proceeding (annotate should handle this) - **Power port direction warnings** — usually benign but review to ensure power ports are connected, not floating

3. Fill in Title Blocks

On each sheet, double-click the title block and fill in:

Field	Sheet 1	Sheet 2	Sheet 3	Sheet 4
Title	Comms — Overview	Comms — Clock Gen	Comms — TX Chain	Comms — RX Chain

Field	Sheet 1	Sheet 2	Sheet 3	Sheet 4
Revision	0.2	0.2	0.2	0.2
Drawn By	Nick Grabbs / K15Y NG	Nick Grabbs / K15Y NG	Nick Grabbs / K15Y NG	Nick Grabbs / K15Y NG
Date	(today)	(today)	(today)	(today)

Field	Sheet 5	Sheet 6	Sheet 7
Title	Comms — Digital Control	Comms — Power	Comms — Connectors
Revision	0.2	0.2	0.2
Drawn By	Nick Grabbs / K15Y NG	Nick Grabbs / K15Y NG	Nick Grabbs / K15Y NG
Date	(today)	(today)	(today)

4. Export PDF

File > Smart PDF (or File > Print > PDF). Save to hardware/comms/releases/Comms_schematic_v0.2.pdf for review.

Complete BOM Summary

Sheet 2: Clock Gen

Ref	Value	Description	Qty
U4	Si5351A-B-GT	Clock generator, MSOP-10	1
Y1	25 MHz, 10pF load	Crystal, HC-49/US or SMD	1
C1	100nF, 16V, X7R	VDD bypass	1
C2	10μF, 10V, X5R	VDD bulk bypass	1
C3	100nF, 16V, X7R	VDDO bypass	1
C4, C5	10pF, 50V, C0G/NP0	Crystal load caps	2
C6, C7	100nF, 16V, X7R	Clock output DC blocks	2
R1, R2	4.7kΩ, 1%	I2C pull-ups	2
R3, R4	33Ω, 1%	Clock output series resistors	2

Sheet 3: TX Chain

Ref	Value	Description	Qty
U1	SN74LVC1G86DBVR	XOR gate, SOT-23-5	1
Q1	BFR92A (or 2N3904 proto)	Tripler transistor, SOT-23	1
U2	ADL5602ARKZ-R7	MMIC gain block, SOT-89	1
C8	100nF, 16V, X7R	XOR VCC bypass	1
C9	10pF, 50V, C0G	Data edge softening	1
C10, C11	100nF, 16V, X7R	Tripler DC blocks (in/out)	2
C12	100nF, 16V, X7R	Tripler supply bypass	1
C13, C16	10pF, 50V, C0G/NP0	Pre-MMIC BPF series caps	2
C14, C15	6.8pF, 50V, C0G/NP0	Pre-MMIC BPF shunt caps	2
C17, C18	100nF, 16V, X7R	MMIC DC blocks (in/out)	2
C19	100nF, 16V, X7R	MMIC supply bypass	1
C20	10μF, 10V, X5R	MMIC supply bulk	1
C45, C48	10pF, 50V, C0G/NP0	Output BPF series caps	2
C46, C47	6.8pF, 50V, C0G/NP0	Output BPF shunt caps	2
R5	100Ω, 1%	Data series resistor	1
R6	47kΩ, 5%	Tripler class-C bias	1
L1	220nH, ±5%, 0805	Tripler RFC	1
L2, L3, L4	15nH, ±2%, 0402	Pre-MMIC BPF inductors	3
L5	220nH, ±5%, 0805	MMIC RFC	1
L6, L7, L8	15nH, ±2%, 0402	Output BPF inductors	3

Sheet 4: RX Chain

Ref	Value	Description	Qty
U3	ADE-1+	Mini-Circuits passive diode mixer, CD636 SMT (6-pad)	1
U10	PSA4-5043+	Mini-Circuits VHF/UHF LNA, SOT-89 (~1 dB NF, ~21 dB gain) — JLC C5240848	1
U7	MCP6022-I/SN	Dual op-amp, SOIC-8 (U7A = filter, U7B = gain)	1
C24	100nF, 16V, X7R	CLK1 DC block (before LO LPF)	1
C28, C29	2.2nF, 25V, C0G/NP0	Sallen-Key filter caps	2

Ref	Value	Description	Qty
C30	100nF, 16V, X7R	Gain stage AC coupling	1
C31	100nF, 16V, X7R	MCP6022 VDD bypass	1
C32	10 μ F, 10V, X5R	Bias point decoupling	1
C49, C52	10pF, 50V, C0G/NP0	2m BPF series caps	2
C50, C51	10pF, 50V, C0G/NP0	2m BPF shunt caps	2
C53	100nF, 16V, X7R	PSA4 input DC block (2m BPF $\rightarrow$ LNA)	1
C54	100nF, 16V, X7R	PSA4 output DC block (LNA $\rightarrow$ mixer RF)	1
C55	100nF, 16V, X7R	PSA4 Vd HF bypass	1
C56	10 μ F, 10V, X5R	PSA4 Vd bulk bypass	1
C57	10pF, 50V, C0G/NP0	LO LPF input shunt cap	1
C58	33pF, 50V, C0G/NP0	LO LPF center shunt cap	1
C59	10pF, 50V, C0G/NP0	LO LPF output shunt cap	1
C60	100nF, 16V, X7R	ADE-1+ IF DC block to Sallen-Key	1
R8, R9	22k Ω , 1%	Sallen-Key filter resistors	2
R10, R11	10k Ω , 1%	Bias voltage divider (1.65V)	2
R12	10k Ω , 1%	Gain stage Rg	1
R13	100k Ω , 1%	Gain stage Rf (gain = 11)	1
R19	51 Ω , 1%	ADE-1+ IF port termination to GND	1
L9, L10, L11	82nH, $\pm$ 2%, 0402	2m BPF inductors	3
L12	1 μ H, wirewound, SRF > 500 MHz	0805/1008 PSA4 Vd bias choke	
L13, L14	68nH, $\pm$ 2%, 0402	LO LPF series inductors	2

Sheet 5: Digital Control

Ref	Value	Description	Qty
Pico	Raspberry Pi Pico	RP2040 module, 2 $\times$ 20 header	1

Ref	Value	Description	Qty
J1	USB Micro-B (optional)	Programming/debug	0–1
J2	1×4 header	SWD debug (optional)	0–1
J3	1×6 header	Spare GPIOs	1
D1–D3	LED, 0603	Status LEDs (optional)	0–3
R14–R16	330Ω–1kΩ	LED resistors (optional)	0–3

Sheet 6: Power

Ref	Value	Description	Qty
D4	SS14 or BAT54S	+5V reverse polarity Schottky	1
D5	BAT54 or SS14	Pico VSYS isolation diode	1
C33	47μF, ≥10V — Panasonic EEE-FK1V470P (wet Al-electrolytic, 35V) or EEH-ZA1V470P (polymer)	+5V input bulk	1
C34	100nF, 16V, X7R	+5V input HF bypass	1
C35	47μF, ≥10V — Panasonic EEE-FK1V470P (wet Al-electrolytic, 35V) or EEH-ZA1V470P (polymer)	+3V3 input bulk	1
C36	100nF, 16V, X7R	+3V3 input HF bypass	1
C37	10μF, 10V, X5R	+3V3 board-level bulk	1
C38	100nF, 16V, X7R	+3V3 board-level HF	1
C39	10μF, 10V, X5R	+5V board-level bulk	1
C40	100nF, 16V, X7R	+5V board-level HF	1
D6, D7	Green LED, 0603	Power indicators (DNP flight)	2
R17, R18	1kΩ	LED resistors (DNP flight)	2

Sheet 7: Connectors

Ref	Value	Description	Qty
J1 (H1 + H2)	2× Samtec ESQ-126-39-G-D	CSKB 2×26 stackthrough socket, 0.1" (2.54 mm), one each for H1 and H2	2
J4	SMA, female, edge-launch	TX antenna connector	1
J5	SMA, female, edge-launch	RX antenna connector	1
J6	1×6 header (optional)	Debug signal header	0–1
C41	10μF, 10V, X5R	+3V3 connector bypass bulk	1
C42	100nF, 16V, X7R	+3V3 connector bypass HF	1
C43	10μF, 10V, X5R	+5V connector bypass bulk	1
C44	100nF, 16V, X7R	+5V connector bypass HF	1
TP1–TP10	Test point pads	Probe points	10

Design Notes Summary

These are the key design decisions. Place a condensed version in the title block “Notes” field or as a text frame on the Top sheet.

1. **Architecture:** 70cm BPSK TX / 2m AFSK RX, single PCB, RP2040 controller
2. **Power:** All regulated power (+3V3, +5V) from EPS via CSKB bus connector. No on-board regulators in stack configuration.
3. **Clock gen:** Si5351A, CLK0=145.67 MHz (TX, tripled to 437 MHz), CLK1=145.9 MHz (RX LO). Both within Si5351A spec (<200 MHz).
4. **BPSK modulation at 145.67 MHz** — XOR gate timing analysis shows 74LVC1G86 fails at 437 MHz but has massive margin at 146 MHz. Phase states preserved through odd-order multiplication.
5. **Frequency tripler:** Class-C biased BFR92A (ft=5 GHz). Bias resistor 47kΩ optimized from breadboard testing (-7.4 dBm @ 437 MHz).
6. **MMIC:** ADL5602 (20 dB gain), replaces SPF5189Z. Expected output ~+13 dBm at 437 MHz.
7. **RX front-end:** PSA4-5043+ LNA (~21 dB gain, ~1 dB NF) feeding ADE-1+ passive double-balanced diode mixer (Mini-Circuits, Level 7, +7 dBm LO). 5-pole Butterworth LPF on the LO line (fc≈200 MHz) suppresses Si5351A square-wave harmonics. 51 Ω resistor terminates the IF port. **Changed from SA612 active mixer** due to SA612 end-of-life sourcing. New architecture has ~5 dB better system NF. See rx_mixer_trade_study.md for full rationale.

8. **Baseband filter:** Sallen-Key LPF, $f_c=3.3$ kHz, -40 dB/decade. LTspice verified: 1200 Hz at -0.56 dB, 2200 Hz at -1.75 dB.
 9. **Gain stage:** MCP6022 non-inverting, gain=11 (20.8 dB), mid-supply bias at 1.65V.
 10. **CSKB connectors:** 2× Samtec ESQ-126-39-G-D (2×26, 0.1" stackthrough), populated at H1 (signals) and H2 (power). Pin assignment follows Pumpkin CubeSat Kit Bus per system/interfaces/cskb_pinmap.md (datasheet Rev. E pp. 13–16). COMMS_IRQ on H1.16, SPI on H1.21/22/23/24, I2C on H1.41/43, +5V on H2.25/26, +3V3 on H2.27/28, DGND on H2.29/30/32, AGND on H2.31, VBAT on H2.45/46.
 11. **RF connectors:** Separate TX/RX SMA, no diplexer for prototype.
 12. **All RF filter caps must be C0G/NP0** — X7R is voltage-dependent and detunes filters at RF.
 13. **I2C bus:** Shared between Si5351A (0x60) and EPS LTC4162 (0x68) — no address conflict. Pull-ups on comms board only (4.7kΩ to 3.3V).
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Related Documents

- Comms overview and design rationale: hardware/comms/design/overview.md
- **RX mixer trade study (SA612 → ADE-1+ + PSA4 LNA):** hardware/comms/design/rx_mixer_trade_study.
- Sallen-Key simulation lab: hardware/comms/design/sallen_key_simulation_lab.md
- Sallen-Key LTspice files: hardware/comms/design/sallen_key_ltspice_lab/
- Si5351A bring-up log: hardware/comms/bringup/si5351a_bringup_log.md
- Tripler breadboard build guide: hardware/comms/bringup/tripler_breadboard.md
- TX test plan: hardware/comms/bringup/tx_test_plan.md
- RX test plan: hardware/comms/bringup/rx_test_plan.md
- EPS Altium schematic guide: hardware/eps/design/altium_eps_schematic.md
- EPS architecture: hardware/eps/design/overview.md
- v0.1 schematic PDF: hardware/comms/design/satellite_comms_draft_v0.1.pdf